MTH251 & MTH252

Mathematical Analysis I & II

Course Textbook

Department of Mathematical Sciences

UNIST

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Course Textbook

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Feel free to open issues or pull requests for corrections.

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Preface

Throughout this textbook, **logical formulas** are interpreted according to the following conventions:

- **Parentheses** have the highest precedence. For example,

  $$(P \lor Q) \land R$$

  is evaluated by first computing $P \lor Q$.

- The **logical connectives** are interpreted in the order

  $$\neg \;>\; \land \;>\; \lor \;>\; \implies \;>\; \iff$$

  Hence

  $$\neg P \land Q \lor R \implies S \iff T$$

  is understood as

  $$((((\neg P) \land Q) \lor R) \implies S) \iff T$$

- A **quantifier** binds the entire formula that follows it. Thus

  $$\forall x, P(x) \land Q(x) \implies R(x)$$

  means

  $$\forall x \, ((P(x) \land Q(x)) \implies R(x))$$

- The symbol $\Downarrow$ is **not** a logical connective; it is used only to indicate the flow of a proof or an explanation. For example,

  $$\begin{gathered} P \\ \Downarrow \\ Q \end{gathered}$$

  merely expresses that P is true, and therefore Q is true in the current discussion; it is not itself a proposition.

Chapter 1

The Number Systems

1.1 The Natural Number Systems

The set $\{1, 2, 3, \ldots\}$ of **natural numbers** is denoted by $\mathbb{N}$, and is defined by the following axioms:

- $1 \in \mathbb{N}$
- $\forall n \in \mathbb{N}, s(n) \in \mathbb{N}$
- $\nexists n \in \mathbb{N}, s(n) = 1$
- $\forall m, n \in \mathbb{N}, s(m) = s(n) \implies m = n$
- If $S \subseteq \mathbb{N}$ satisfies
  - $1 \in S$
  - $\forall n \in S, s(n) \in S$

  then $S = \mathbb{N}$

where $s(n)$ is called the **successor** of n. These axioms are called the **Peano axioms** (or *Peano Postulates*).

Remark. The **Peano axioms** do not guarantee the *existence* of $\mathbb{N}$, which is guaranteed by the **axiom of infinity** in the **ZFC axioms**. In this textbook, the **ZFC axioms** is not discussed, so we will just assume that $\mathbb{N}$ exists.

Definition 1.1.1. Let $P(n)$ be a statement about $n \in \mathbb{N}$. Define

$$S \coloneqq \{n \in \mathbb{N} \mid P(n) \text{ is true}\} \subseteq \mathbb{N}$$

If $P(n)$ satisfies

- $P(1)$ is true
- $\forall n \in \mathbb{N}, P(n)$ is true $\implies P(s(n))$ is true

Then S satisfies

- $1 \in S$
- $\forall n \in S, s(n) \in S$

Therefore,

$$\begin{gathered} S = \mathbb{N} \\ \Downarrow \\ \forall n \in \mathbb{N}, P(n) \text{ is true} \end{gathered}$$

This method of proof is called the **principle of mathematical induction**.

Definition 1.1.2. Let $n, m \in \mathbb{N}$ be given.

- The **addition** $+$ on $\mathbb{N}$ is defined as
  - $n + 1 := s(n)$
  - $n + s(m) := s(n + m)$
- The **multiplication** $\cdot$ on $\mathbb{N}$ is defined as
  - $n \cdot 1 := n$
  - $n \cdot s(m) := n \cdot m + n$

Remark. Strictly speaking, Definition 1.1.2 requires the **recursion theorem** to be strictly defined. In this textbook, the **recursion theorem** is not discussed, so we will just assume that the operations on $\mathbb{N}$ are **well-defined**.

Theorem 1.1.3. *Let $n, m, k \in \mathbb{N}$ be given. Then the following properties hold:*

- $n + m = m + n$
- $(n + m) + k = n + (m + k)$
- $n \cdot m = m \cdot n$
- $(n \cdot m) \cdot k = n \cdot (m \cdot k)$
- $n \cdot (m + k) = n \cdot m + n \cdot k$

Proof. We will not discuss the proof of this theorem, since it relies on the recursive definitions. $\square$

Theorem 1.1.4. $\mathbb{N}$ *has the **well-ordering property**, which states that*

$$\forall S \subseteq \mathbb{N}, S \neq \varnothing \implies \exists m \in S, m = \min S$$

Proof. Assume that

$$\exists S \subseteq \mathbb{N}, S \neq \varnothing \wedge (\nexists m \in S, m = \min S)$$

Define

- $T \coloneqq \mathbb{N} \setminus S \subseteq \mathbb{N}$
- $L \coloneqq \{n \in \mathbb{N} \mid \{1, 2, \ldots, n\} \subseteq T\} \subseteq \mathbb{N}$

We will show that

$$\begin{gathered} L = \mathbb{N} \\ \Downarrow \\ T = \mathbb{N} \\ \Downarrow \\ S = \varnothing \\ \Downarrow \\ \perp \end{gathered}$$

- **Base case:** If $1 \in S$, then

  $$\begin{gathered} 1 \in S \\ \Downarrow \\ 1 = \min S \\ \Downarrow \\ \perp \end{gathered}$$

  Therefore,

  $$\begin{gathered} 1 \notin S \\ \Downarrow \\ 1 \in T \\ \Downarrow \\ 1 \in L \end{gathered}$$

- **Inductive step:** Let $n \in L$ be given. Then

  $$\begin{gathered} n \in L \\ \Downarrow \\ \{1, 2, \ldots, n\} \subseteq T \\ \Downarrow \\ \{1, 2, \ldots, n\} \cap S = \varnothing \end{gathered}$$

  If $n + 1 \in S$, then

  $$\begin{gathered} n + 1 \in S \\ \Downarrow \\ n + 1 = \min S \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$\begin{gathered} n+1 \notin S \\ \Downarrow \\ n+1 \in T \\ \Downarrow \\ \{1,2,\ldots,n+1\} \subseteq T \\ \Downarrow \\ n+1 \in L \end{gathered}$$

By the *principle of mathematical induction*, $L = \mathbb{N}$. □

1.2 Relations

Let S be a set. A **relation** $\sim$ on S is a subset of $S \times S$. Equivalently, a subset R of $S \times S$ is a relation on S. That is, for every $x, y \in S$,

- $R = \{(x,y) \in S \times S \mid x \sim y\}$
- $x \sim y \iff (x,y) \in R$

Definition 1.2.1. Let S be a set. A relation $\sim$ on S is called an **equivalence relation** if it satisfies the following properties:

- **Reflexivity**: $\forall x \in S, x \sim x$
- **Symmetry**: $\forall x, y \in S, x \sim y \implies y \sim x$
- **Transitivity**: $\forall x, y, z \in S, x \sim y \wedge y \sim z \implies x \sim z$

Definition 1.2.2. Let S be a set. A collection $\{A_\alpha\}_{\alpha \in I}$ of sets is called a **partition** of S if

- $\forall \alpha \in I, A_\alpha \neq \varnothing \wedge A_\alpha \subseteq S$
- $\forall \alpha, \beta \in I, \alpha \neq \beta \implies A_\alpha \cap A_\beta = \varnothing$
- $\displaystyle\bigcup_{\alpha \in I} A_\alpha = S$

Definition 1.2.3. Let S be a set, and let $\sim$ be an equivalence relation on S. For each $x \in S$, the **equivalence class** of x is defined as

$$[x] \coloneqq \{y \in S \mid y \sim x\}$$

We denote the set of all equivalence classes of S by

$$S/\sim \coloneqq \{[x] \mid x \in S\}$$

Theorem 1.2.4. *Let S be a set, and let $\sim$ be an equivalence relation on S. Then*

$$\forall x, y \in S, [x] \cap [y] \neq \varnothing \implies [x] = [y]$$

Proof. If $[x] \cap [y] \neq \varnothing$, then

$$\begin{gathered} \exists z \in S, z \in [x] \cap [y] \\ \Downarrow \\ z \in [x] \wedge z \in [y] \\ \Downarrow \\ z \sim x \wedge z \sim y \\ \Downarrow \\ x \sim z \wedge z \sim y \\ \Downarrow \\ x \sim y \end{gathered}$$

Therefore, for every $w \in [x]$,

$$\begin{gathered} w \sim x \wedge x \sim y \\ \Downarrow \\ w \sim y \\ \Downarrow \\ w \in [y] \end{gathered}$$

Therefore,

$$[x] \subseteq [y]$$

Similarly,

$$[y] \subseteq [x]$$

□

Theorem 1.2.5. *Let S be a set, and let $\{A_\alpha\}_{\alpha \in I}$ be a partition of S. Define*

$$x \sim y \iff \exists \alpha \in I, x \in A_\alpha \wedge y \in A_\alpha$$

for all $x, y \in S$. Then

(A) *$\sim$ is an equivalence relation on S*

(B) *$\{A_\alpha\}_{\alpha \in I} = S/\sim$*

Proof. **(A)** Let $x, y, z \in S$ be given.

- **Reflexivity**: Since $x \in S$,

$$\begin{gathered} x \in S \\ \Downarrow \\ x \in \bigcup_{\alpha \in I} A_\alpha \\ \Downarrow \\ \exists \alpha \in I, x \in A_\alpha \\ \Downarrow \\ x \sim x \end{gathered}$$

- **Symmetry**: If $x \sim y$, then

$$\begin{gathered} \exists \alpha \in I, x \in A_\alpha \wedge y \in A_\alpha \\ \Downarrow \\ y \in A_\alpha \wedge x \in A_\alpha \\ \Downarrow \\ y \sim x \end{gathered}$$

- **Transitivity**: If $x \sim y \wedge y \sim z$, then
  - $\exists \alpha \in I, x \in A_\alpha \wedge y \in A_\alpha$
  - $\exists \beta \in I, y \in A_\beta \wedge z \in A_\beta$

  Therefore,

$$\begin{gathered} y \in A_\alpha \wedge y \in A_\beta \\ \Downarrow \\ A_\alpha \cap A_\beta \neq \varnothing \\ \Downarrow \\ \alpha = \beta \\ \Downarrow \\ x \in A_\alpha \wedge z \in A_\alpha \\ \Downarrow \\ x \sim z \end{gathered}$$

(B) Let $x \in S$ be given. Then

$$\begin{gathered} x \in S \\ \Downarrow \\ x \in \bigcup_{\alpha \in I} A_\alpha \\ \Downarrow \\ \exists \alpha \in I, x \in A_\alpha \end{gathered}$$

We will show that $A_\alpha = [x]$.

- ($\subseteq$) Let $y \in A_\alpha$ be given. Then

$$\begin{gathered} y \in A_\alpha \wedge x \in A_\alpha \\ \Downarrow \\ y \sim x \\ \Downarrow \\ y \in [x] \end{gathered}$$

- ($\supseteq$) Let $y \in [x]$ be given. Then

$$\begin{gathered} y \sim x \\ \Downarrow \\ \exists \beta \in I, y \in A_\beta \wedge x \in A_\beta \\ \Downarrow \\ x \in A_\alpha \wedge x \in A_\beta \\ \Downarrow \\ x \in A_\alpha \cap A_\beta \\ \Downarrow \\ A_\alpha \cap A_\beta \neq \varnothing \\ \Downarrow \\ \alpha = \beta \\ \Downarrow \\ y \in A_\alpha \end{gathered}$$

To sum up,

$$\begin{gathered} \forall x \in S, \exists \alpha \in I, A_\alpha = [x] \\ \Downarrow \\ S/\sim \, \subseteq \{A_\alpha\}_{\alpha \in I} \end{gathered}$$

Also,

$$\begin{gathered} \forall \alpha \in I, A_\alpha \neq \varnothing \\ \Downarrow \\ \exists x \in A_\alpha \subseteq S \\ \Downarrow \\ A_\alpha = [x] \\ \Downarrow \\ \{A_\alpha\}_{\alpha \in I} \subseteq S/\sim \end{gathered}$$

Therefore,

$$\{A_\alpha\}_{\alpha \in I} = S/\sim$$

$\square$

1.3 The Integer and Rational Number Systems

Define a relation $\sim$ on $\mathbb{N} \times \mathbb{N}$ as

$$(a, b) \sim (c, d) \iff a + d = b + c$$

Let $(a, b), (c, d), (e, f) \in \mathbb{N} \times \mathbb{N}$ be given. By Theorem 1.1.3,

- **Reflexivity**: Since $a \in \mathbb{N} \wedge b \in \mathbb{N}$,

$$\begin{gathered} a + b = b + a \\ \Downarrow \\ (a, b) \sim (a, b) \end{gathered}$$

- **Symmetry**: If $(a, b) \sim (c, d)$, then

$$\begin{gathered} a + d = b + c \\ \Downarrow \\ c + b = d + a \\ \Downarrow \\ (c, d) \sim (a, b) \end{gathered}$$

- **Transitivity**: If $(a, b) \sim (c, d) \wedge (c, d) \sim (e, f)$, then

$$\begin{gathered} a + d = b + c \wedge c + f = d + e \\ \Downarrow \\ a + d + f = b + c + f = b + d + e \\ \Downarrow \\ a + f = b + e \\ \Downarrow \\ (a, b) \sim (e, f) \end{gathered}$$

Therefore, $\sim$ is an equivalence relation on $\mathbb{N} \times \mathbb{N}$. The set of **integers** $\mathbb{Z}$ is defined as

$$\mathbb{Z} := (\mathbb{N} \times \mathbb{N}) / \sim$$

Definition 1.3.1. Let $[(a, b)], [(c, d)] \in \mathbb{Z}$ be given.

- The **addition** $+$ on $\mathbb{Z}$ is defined as

$$[(a, b)] + [(c, d)] := [(a + c, b + d)]$$

- The **multiplication** $\cdot$ on $\mathbb{Z}$ is defined as

$$[(a, b)] \cdot [(c, d)] := [(a \cdot c + b \cdot d, a \cdot d + b \cdot c)]$$

Remark. The operations $+$ and $\cdot$ are **well-defined** on $\mathbb{Z}$; that is, they do not depend on the **choice of representatives** of the equivalence classes. Indeed, if

- $(a, b) \sim (a', b')$

- $(c,d) \sim (c',d')$

then one can verify that

- $(a+c, b+d) \sim (a'+c', b'+d')$
- $(a \cdot c + b \cdot d, a \cdot d + b \cdot c) \sim (a' \cdot c' + b' \cdot d', a' \cdot d' + b' \cdot c')$

The same issue arises for **addition** and **multiplication** on $\mathbb{Q}$, whose **well-definedness** could be verified in exactly the same manner.

Theorem 1.3.2. *Let $n, m, k \in \mathbb{Z}$ be given. Then the following properties hold:*

- $n + m = m + n$
- $(n+m) + k = n + (m+k)$
- $n \cdot m = m \cdot n$
- $(n \cdot m) \cdot k = n \cdot (m \cdot k)$
- $n \cdot (m+k) = (n \cdot m) + (n \cdot k)$

Proof. The proof follows directly from Definition 1.3.1 and theorem 1.1.3. □

Define
$$\mathbb{Z}^+ := \{[(n+1,1)] \in \mathbb{Z} \mid n \in \mathbb{N}\} \subseteq \mathbb{Z}$$
Define a relation $\sim$ on $\mathbb{Z} \times \mathbb{Z}^+$ as
$$(a,b) \sim (c,d) \iff a \cdot d = b \cdot c$$
Let $(a,b), (c,d), (e,f) \in \mathbb{Z} \times \mathbb{Z}^+$ be given. By Theorem 1.3.2,

- **Reflexivity**: Since $a \in \mathbb{Z} \wedge b \in \mathbb{Z}^+$,
$$\begin{gathered} a \cdot b = b \cdot a \\ \Downarrow \\ (a,b) \sim (a,b) \end{gathered}$$
- **Symmetry**: If $(a,b) \sim (c,d)$, then
$$\begin{gathered} a \cdot d = b \cdot c \\ \Downarrow \\ c \cdot b = d \cdot a \\ \Downarrow \\ (c,d) \sim (a,b) \end{gathered}$$

- **Transitivity**: If $(a, b) \sim (c, d) \wedge (c, d) \sim (e, f)$, then

$$\begin{gathered} a \cdot d = b \cdot c \wedge c \cdot f = d \cdot e \\ \Downarrow \\ a \cdot d \cdot f = b \cdot c \cdot f = b \cdot d \cdot e \\ \Downarrow \\ a \cdot f = b \cdot e \\ \Downarrow \\ (a, b) \sim (e, f) \end{gathered}$$

Therefore, $\sim$ is an equivalence relation on $\mathbb{Z} \times \mathbb{Z}^+$. The set of **rational numbers** $\mathbb{Q}$ is defined as

$$\mathbb{Q} \coloneqq (\mathbb{Z} \times \mathbb{Z}^+)/\sim$$

Definition 1.3.3. Let $[(a, b)], [(c, d)] \in \mathbb{Q}$ be given.

- The **addition** $+$ on $\mathbb{Q}$ is defined as

$$[(a, b)] + [(c, d)] \coloneqq [(ad + bc, bd)]$$

- The **multiplication** $\cdot$ on $\mathbb{Q}$ is defined as

$$[(a, b)] \cdot [(c, d)] \coloneqq [(ac, bd)]$$

Theorem 1.3.4. *Let $p, q, r \in \mathbb{Q}$ be given. Then the following properties hold:*

- $p + q = q + p$
- $(p + q) + r = p + (q + r)$
- $p \cdot q = q \cdot p$
- $(p \cdot q) \cdot r = p \cdot (q \cdot r)$
- $p \cdot (q + r) = (p \cdot q) + (p \cdot r)$

Proof. The proof follows directly from Definition 1.3.3 and theorem 1.3.2. □

Remark. Strictly speaking, the constructions of $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$ produce **different kind** of sets. For example,

- $n \in \mathbb{N}$
- $[(n + 1, 1)] \in \mathbb{Z}$

- $[(n, [(2, 1)])] \in \mathbb{Q}$

are distinct objects in the underlying set-theoretic construction. Nevertheless, there are **canonical embeddings**

- $\mathbb{N} \xrightarrow{\cong} \mathbb{Z}$ such that $n \mapsto [(n + 1, 1)]$
- $\mathbb{Z} \xrightarrow{\cong} \mathbb{Q}$ such that $a \mapsto [(a, [(2, 1)])]$

which preserve the **algebraic operations**. We therefore identify each number system with its image in the next larger one and write

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q}$$

This identification is justified by the fact that the embeddings are canonical, requiring **no arbitrary choices**. Throughout this textbook, inclusions such as

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$$

should be understood as canonical identifications rather than literal inclusions arising directly from the set-theoretic constructions.

Remark. Define

$$0 \coloneqq [(1, 1)] \in \mathbb{Z}$$

In this textbook, we will use the following notations:

- $\mathbb{N}_0 \coloneqq \mathbb{N} \cup \{0\} \subseteq \mathbb{Z}$
- $\mathbb{N}_n \coloneqq \{k \in \mathbb{N} \mid k \geq n\} = \{n, n + 1, n + 2, \dots\} \subseteq \mathbb{N}$ for each $n \in \mathbb{N}$
- $\mathbb{N}^0 \coloneqq \varnothing$
- $\mathbb{N}^m \coloneqq \{k \in \mathbb{N} \mid k \leq m\} = \{1, 2, \dots, m\} \subseteq \mathbb{N}$ for each $m \in \mathbb{N}$
- $\mathbb{N}_n^m \coloneqq \{k \in \mathbb{N}_n \mid k \leq m\} = \{n, n + 1, \dots, m\} \subseteq \mathbb{N}_0$ for each $n, m \in \mathbb{N}_0$ with $n \leq m$

1.4 Ordered Sets

A set S is called an **ordered set** if it is equipped with a relation $<$ such that for every $x, y, z \in S$, the following properties hold:

- $x < y \land y < z \implies x < z$
- Only one of the relations $x < y$, $x = y$, or $x > y$ holds

The relation $<$ is called an **(total) order** on S.

Definition 1.4.1. The relations $>$, $\leq$, and $\geq$ are defined as

- $x > y \iff y < x$
- $x \geq y \iff x > y \vee x = y$
- $x \leq y \iff y \geq x$

Example 1.4.2. $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ are ordered sets with the following relations:

- $\forall n, m \in \mathbb{N}, m < n \iff \exists k \in \mathbb{N}, n = m + k$
- $\forall n, m \in \mathbb{Z}, m < n \iff \exists k \in \mathbb{Z}^+, n = m + k$
- $\forall p, q \in \mathbb{Q}, q < p \iff \exists r \in \mathbb{Q}^+, p = q + r$

where

$$\mathbb{Q}^+ \coloneqq \left\{ [(p,q)] \in \mathbb{Q} \;\middle|\; p, q \in \mathbb{Z}^+ \right\} \subseteq \mathbb{Q}$$

Definition 1.4.3. Let S be an ordered set and $E \subseteq S$.

- E is said to be **bounded above** if

$$\exists M \in S, \forall x \in E, x \leq M$$

  Such an M is called an **upper bound** of E.

- E is said to be **bounded below** if

$$\exists m \in S, \forall x \in E, m \leq x$$

  Such an m is called a **lower bound** of E.

Theorem 1.4.4. *Let $E \subseteq \mathbb{Z}$ be nonempty. Then*

(A) *E is bounded below $\implies$ E has a least element*

(B) *E is bounded above $\implies$ E has a greatest element*

Proof. **(A)** Let E be bounded below. Then,

$$\begin{gathered} E \text{ is bounded below} \\ \Downarrow \\ \exists m \in \mathbb{Z}, \forall x \in E, m \leq x \end{gathered}$$

Define

$$F \coloneqq \{x - m + 1 \mid x \in E\} \subseteq \mathbb{N}$$

Then

$$\begin{gathered} E \neq \varnothing \\ \Downarrow \\ F \neq \varnothing \end{gathered}$$

By Theorem 1.1.4,

$$\begin{gathered} \exists n \in F, n = \min F \\ \Downarrow \\ \forall x \in E, n \leq x - m + 1 \\ \Downarrow \\ \forall x \in E, n + m - 1 \leq x \end{gathered}$$

Since $n = \min F$,

$$\begin{gathered} n \in F \\ \Downarrow \\ \exists x_0 \in E, n = x_0 - m + 1 \\ \Downarrow \\ x_0 = n + m - 1 = \min E \end{gathered}$$

(B) Let E be bounded above. Then,

$$\begin{gathered} E \text{ is bounded above} \\ \Downarrow \\ \exists M \in \mathbb{Z}, \forall x \in E, x \leq M \end{gathered}$$

Define

$$F := \{-x + M + 1 \mid x \in E\} \subseteq \mathbb{N}$$

Then

$$\begin{gathered} E \neq \varnothing \\ \Downarrow \\ F \neq \varnothing \end{gathered}$$

By Theorem 1.1.4,

$$\begin{gathered} \exists n \in F, n = \min F \\ \Downarrow \\ \forall x \in E, n \leq -x + M + 1 \\ \Downarrow \\ \forall x \in E, x \leq M - n + 1 \end{gathered}$$

Since $n = \min F$,

$$\begin{gathered} n \in F \\ \Downarrow \\ \exists x_0 \in E, n = -x_0 + M + 1 \\ \Downarrow \\ x_0 = M - n + 1 = \max E \end{gathered}$$

$\square$

Definition 1.4.5. Let S be an ordered set, and let $E \subseteq S$ be bounded above. An element $\alpha \in S$ is called a **supremum** (or *least upper bound*) of E if

- α is an upper bound of E
- $\forall \beta \in S, \beta < \alpha \implies \beta$ is not an upper bound of E

A supremum of E is denoted by

$$\alpha = \sup E$$

Similarly, let $E \subseteq S$ be bounded below. An element $\alpha \in S$ is called an **infimum** (or *greatest lower bound*) of E if

- α is a lower bound of E
- $\forall \beta \in S, \beta > \alpha \implies \beta$ is not a lower bound of E

An infimum of E is denoted by

$$\alpha = \inf E$$

Example 1.4.6. Define

$$S \coloneqq \left\{ p \in \mathbb{Q}^+ \;\middle|\; p^2 < 2 \right\}$$

- Let $p \in S$ be given. Define

  $$q \coloneqq p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2} > p$$

  Then

  $$\begin{gathered} q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2} < 0 \\ \Downarrow \\ q \in S \end{gathered}$$

  This implies that $p \in S$ is not an upper bound of S.

- Let $p \in \mathbb{Q}^+ \setminus S$ be given.

  $$\nexists q \in \mathbb{Q}^+, q^2 = 2$$

  Therefore,

  $$\begin{gathered} p^2 \geq 2 \\ \Downarrow \\ p^2 > 2 \\ \Downarrow \\ \forall x \in S, x^2 < 2 < p^2 \\ \Downarrow \\ \forall x \in S, x < p \\ \Downarrow \\ p \text{ is an upper bound of } S \end{gathered}$$

Define

$$q \coloneqq p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2} < p$$

Then

$$\begin{gathered} q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2} > 0 \\ \Downarrow \\ q^2 > 2 \\ \Downarrow \\ \forall x \in S, x^2 < 2 < q^2 \\ \Downarrow \\ \forall x \in S, x < q \\ \Downarrow \\ q \text{ is an upper bound of } S \end{gathered}$$

This implies that p is not a least upper bound of S.

To sum up,

$$\nexists \alpha \in \mathbb{Q}^+, \alpha = \sup S$$

Definition 1.4.7. Let S be an ordered set and $E \subseteq S$. If

- $E \neq \varnothing$
- E is bounded above

implies that

$$\exists \alpha \in S, \alpha = \sup E$$

then S is said to have the **least upper bound property**.

Theorem 1.4.8. *Let S be an ordered set and $E \subseteq S$. Then*

- *S has the least upper bound property*
- *$E \neq \varnothing$*
- *E is bounded below*

implies that

$$\exists \alpha \in S, \alpha = \inf E$$

Proof. Assume that

- S has the *least upper bound property*
- $E \neq \varnothing$

- E is bounded below

Define

$$L := \{\lambda \in S \mid \forall x \in E, \lambda \leq x\} \subseteq S$$

Then

$$\begin{gathered} E \text{ is bounded below} \\ \Downarrow \\ L \neq \varnothing \end{gathered}$$

Also,

$$\begin{gathered} E \neq \varnothing \\ \Downarrow \\ \exists \beta \in E \\ \Downarrow \\ \forall \lambda \in L, \lambda \leq \beta \\ \Downarrow \\ L \text{ is bounded above} \end{gathered}$$

By the *least upper bound property*,

$$\exists \alpha \in S, \alpha = \sup L$$

We will show that

$$\alpha = \inf E$$

Assume that $\alpha \notin L$. Then

$$\begin{gathered} \alpha \notin L \\ \Downarrow \\ \exists \beta \in E, \beta < \alpha \end{gathered}$$

Since $\alpha = \sup L$,

$$\begin{gathered} \alpha = \sup L \\ \Downarrow \\ \beta \text{ is not an upper bound of } L \\ \Downarrow \\ \exists \gamma \in L, \beta < \gamma \leq \alpha \end{gathered}$$

Then

$$\begin{gathered} \gamma \in L \\ \Downarrow \\ \forall x \in E, \gamma \leq x \\ \Downarrow \\ \gamma \leq \beta \\ \Downarrow \\ \beta < \gamma \leq \beta \\ \Downarrow \\ \bot \end{gathered}$$

Therefore,

$$\alpha \in L$$
$$\Downarrow$$
$$\alpha \text{ is a lower bound of } E$$

Since $\alpha = \sup L$,

$$\forall \delta \in S, \delta > \alpha \implies \delta \notin L$$

To sum up,

- α is a lower bound of E
- $\forall \delta \in S, \delta > \alpha \implies \delta$ is not a lower bound of E

Therefore,

$$\alpha = \inf E$$

$\square$

1.5 Fields and Ordered Fields

A set F is called a **field** if it is equipped with two binary operations

- **addition** $+$
- **multiplication** $\cdot$

and satisfies the following axioms:

(A) Axioms for **addition**

- $\forall x, y \in F, x + y \in F$
- $\forall x, y \in F, x + y = y + x$
- $\forall x, y, z \in F, (x + y) + z = x + (y + z)$
- $\exists 0 \in F, \forall x \in F, x + 0 = x$
- $\forall x \in F, \exists - x \in F, x + (-x) = 0$

(B) Axioms for **multiplication**

- $\forall x, y \in F, x \cdot y \in F$
- $\forall x, y \in F, x \cdot y = y \cdot x$
- $\forall x, y, z \in F, (x \cdot y) \cdot z = x \cdot (y \cdot z)$
- $\exists 1 \in F, 1 \neq 0 \wedge (\forall x \in F, x \cdot 1 = x)$
- $\forall x \in F, x \neq 0 \implies \exists x^{-1} \in F, x \cdot x^{-1} = 1$

(C) The **distributive law**

- $\forall x, y, z \in F, x \cdot (y + z) = x \cdot y + x \cdot z$

A set F is called an **ordered field** if

- F is a field
- F is an ordered set

and satisfies the following axioms:

- $\forall x, y, z \in F, x < y \implies x + z < y + z$
- $\forall x, y \in F, x > 0 \wedge y > 0 \implies x \cdot y > 0$

Example 1.5.1. $(\mathbb{Q}, +, \cdot)$ is an ordered field, while $(\mathbb{Z}, +, \cdot)$ is not an ordered field since

$$\nexists 2^{-1} \in \mathbb{Z}, 2 \cdot 2^{-1} = 1$$

1.6 The Real Number System

An ordered field $(\mathbb{R}, +, \cdot)$ has the *least upper bound property*. The construction of $\mathbb{R}$ from $\mathbb{Q}$ is rather complicated, so it will be discussed separately in Appendix. In this section, we will just assume that $\mathbb{R}$ exists.

Theorem 1.6.1. *$\mathbb{R}$ has the **Archimedean property**, which states that*

$$\forall x \in \mathbb{R}^+, \forall y \in \mathbb{R}, \exists n \in \mathbb{N}, nx > y$$

where

$$\mathbb{R}^+ := \{x \in \mathbb{R} \mid x > 0\}$$

Proof. Define

$$E := \{nx \mid nx \leq y,\ n \in \mathbb{N}\} \subseteq \mathbb{R}$$

If $E = \varnothing$, then

$$\begin{gathered} 1 \in \mathbb{N} \wedge 1 \cdot x \notin E \\ \Downarrow \\ 1 \cdot x > y \end{gathered}$$

Otherwise,

$$E \neq \varnothing$$

E is bounded above by y. By the *least upper bound property*,

$$\exists \alpha \in \mathbb{R}, \alpha = \sup E$$

Since $\alpha = \sup E$,

$$\begin{gathered}\alpha - x < \alpha \\ \Downarrow \\ \alpha - x \text{ is not an upper bound of } E \\ \Downarrow \\ \exists m \in \mathbb{N}, mx > \alpha - x \\ \Downarrow \\ (m+1)x > \alpha \\ \Downarrow \\ m+1 \in \mathbb{N} \wedge (m+1)x \notin E \\ \Downarrow \\ (m+1)x > y\end{gathered}$$

Therefore, we can take $n = m + 1$ to complete the proof. □

Corollary 1.6.2.

$$\forall \varepsilon \in \mathbb{R}^+, \forall b \in (1, +\infty), \exists n \in \mathbb{N}, b^{-n} < \varepsilon$$

Proof. The **Bernoulli's inequality** states that

$$\forall x \in [-1, +\infty), \forall n \in \mathbb{N}, (1+x)^n \geq 1 + nx$$

By Theorem 1.6.1,

$$\begin{gathered}\forall a \in \mathbb{R}^+, \exists n \in \mathbb{N}, na > \frac{1}{\varepsilon} \\ \Downarrow \\ (1+a)^n \geq 1 + na > \frac{1}{\varepsilon} \implies (1+a)^{-n} < \varepsilon\end{gathered}$$

Then

$$\begin{gathered}b \in (1, +\infty) \\ \Downarrow \\ b - 1 \in \mathbb{R}^+ \\ \Downarrow \\ \exists n \in \mathbb{N}, n(b-1) > \frac{1}{\varepsilon} \\ \Downarrow \\ (1 + (b-1))^n \geq 1 + n(b-1) > \frac{1}{\varepsilon} \implies (1 + (b-1))^{-n} < \varepsilon \\ \Downarrow \\ b^{-n} < \varepsilon\end{gathered}$$

□

Theorem 1.6.3. *Let $E \subseteq \mathbb{Z}$ be nonempty. Then*

(A) *E is bounded below in $\mathbb{R} \implies E$ has a least element*

(B) *E is bounded above in $\mathbb{R} \implies E$ has a greatest element*

Proof. **(A)** Let E be bounded below in $\mathbb{R}$. Then

$$\exists \alpha \in \mathbb{R}, \forall x \in E, \alpha \leq x$$

If $\alpha \geq 0$, then

$$\begin{gathered} \forall x \in E, 0 \leq \alpha \leq x \\ \Downarrow \\ 0 \in \mathbb{Z} \text{ is a lower bound of } E \end{gathered}$$

Otherwise, by Theorem 1.6.1,

$$\begin{gathered} -\alpha \in \mathbb{R} \\ \Downarrow \\ \exists n \in \mathbb{N}, n > -\alpha \\ \Downarrow \\ \forall x \in E, -n < \alpha \leq x \\ \Downarrow \\ -n \in \mathbb{Z} \text{ is a lower bound of } E \end{gathered}$$

Therefore, E is bounded below in $\mathbb{Z}$. By Theorem 1.4.4, E has a least element.

(B) Let E be bounded above in $\mathbb{R}$. Then

$$\exists \alpha \in \mathbb{R}, \forall x \in E, x \leq \alpha$$

Then

$$\begin{gathered} \alpha \in \mathbb{R} \\ \Downarrow \\ \exists n \in \mathbb{N}, n > \alpha \\ \Downarrow \\ \forall x \in E, x \leq \alpha < n \\ \Downarrow \\ n \in \mathbb{Z} \text{ is an upper bound of } E \end{gathered}$$

Therefore, E is bounded above in $\mathbb{Z}$. By Theorem 1.4.4, E has a greatest element.

□

Definition 1.6.4. Let $x \in \mathbb{R}$ be given. The **floor function** $\lfloor x \rfloor$ and **ceiling function** $\lceil x \rceil$ are defined as

- $\lfloor x \rfloor := \max\{k \in \mathbb{Z} \mid k \leq x\}$

- $\lceil x \rceil := \min\{k \in \mathbb{Z} \mid k \geq x\}$

Theorem 1.6.3 guarantees the *existence* of $\lfloor x \rfloor$ and $\lceil x \rceil$.

Theorem 1.6.5. *$\mathbb{Q}$ is **dense** in $\mathbb{R}$, which states that*

$$\forall x, y \in \mathbb{R}, x < y \implies \exists p \in \mathbb{Q}, x < p < y$$

Proof. Define

$$\varepsilon := y - x > 0$$

By Theorem 1.6.1,

$$\exists n \in \mathbb{N}, \frac{1}{n} < \varepsilon$$

Define

$$S := \left\{ k \in \mathbb{Z} \;\middle|\; \frac{k}{n} \leq x \right\} \subseteq \mathbb{Z}$$

Then

$$\begin{gathered} \lfloor x \rfloor \leq x < \lfloor x \rfloor + 1 \\ \Downarrow \\ n\lfloor x \rfloor \leq nx < n\lfloor x \rfloor + n \\ \Downarrow \\ n\lfloor x \rfloor \in S \\ \Downarrow \\ S \neq \varnothing \end{gathered}$$

S is bounded above by nx. By Theorem 1.6.3,

$$\exists m \in \mathbb{Z}, m = \max S$$

Since $m = \max S$,

$$\begin{gathered} m \in S \\ \Downarrow \\ \frac{m}{n} \leq x \\ \Downarrow \\ \frac{m+1}{n} \leq x + \frac{1}{n} \end{gathered}$$

Also,

$$\begin{gathered} m + 1 \notin S \\ \Downarrow \\ \frac{m+1}{n} > x \end{gathered}$$

To sum up,

$$x < \frac{m+1}{n} \leq x + \frac{1}{n} < x + \varepsilon = x + (y - x) = y$$

Therefore, we can take $p = \dfrac{m+1}{n} \in \mathbb{Q}$ to complete the proof. $\square$

Theorem 1.6.6.

$$\forall x \in \mathbb{R}^+, \forall n \in \mathbb{N}, \exists! y \in \mathbb{R}^+, y^n = x$$

Such y is denoted by

- $y = \sqrt[n]{x}$
- $y = x^{\frac{1}{n}}$

Proof. Define

$$E := \left\{ t \in \mathbb{R}^+ \;\middle|\; t^n < x \right\}$$

If $x \geq 1$, then

$$\begin{gathered} \forall m \in \mathbb{N}, 2^{-m} < 1 \leq x \\ \Downarrow \\ 2^{-1} \in E \\ \Downarrow \\ E \neq \varnothing \end{gathered}$$

Otherwise,

$$\begin{gathered} x < 1 \\ \Downarrow \\ \forall m \in \mathbb{N}, x^{2m} < x \\ \Downarrow \\ x^2 \in E \\ \Downarrow \\ E \neq \varnothing \end{gathered}$$

E is bounded above by max$\{x, 1\}$. By the *least upper bound property*,

$$\exists \alpha \in \mathbb{R}^+, \alpha = \sup E$$

We will show that $\alpha^n = x$. Let $0 < a < b < \infty$ be given. Then

$$b^n - a^n = (b - a) \sum_{k=0}^{n-1} b^{n-1-k} a^k < (b - a) n b^{n-1}$$

Assume that $\alpha^n < x$. By Theorem 1.6.5,

$$\exists h \in \mathbb{R}^+, h < \min \left\{ 1, \frac{x - \alpha^n}{n(\alpha + 1)^{n-1}} \right\}$$

Put $a = \alpha$ and $b = \alpha + h$. Then

$$\begin{gathered} (\alpha + h)^n - \alpha^n < hn(\alpha + h)^{n-1} < hn(\alpha + 1)^{n-1} \\ \Downarrow \\ (\alpha + h)^n < \alpha^n + hn(\alpha + 1)^{n-1} < \alpha^n + (x - \alpha^n) = x \end{gathered}$$

Since $\alpha = \sup E$,

$$\begin{gathered}(\alpha+h)^n < x \\ \Downarrow \\ \alpha + h \in E \\ \Downarrow \\ \perp\end{gathered}$$

Assume that $\alpha^n > x$. By Theorem 1.6.5,

$$\exists h \in \mathbb{R}^+, h < \min\left\{\alpha, \frac{\alpha^n - x}{n\alpha^{n-1}}\right\}$$

Put $a = \alpha - h$ and $b = \alpha$. Then

$$\begin{gathered}\alpha^n - (\alpha-h)^n < hn\alpha^{n-1} \\ \Downarrow \\ (\alpha-h)^n > \alpha^n - hn\alpha^{n-1} > \alpha^n - (\alpha^n - x) = x\end{gathered}$$

Since $\alpha = \sup E$,

$$\begin{gathered}(\alpha-h)^n > x \\ \Downarrow \\ \forall t \in E, t^n < x < (\alpha-h)^n \\ \Downarrow \\ \forall t \in E, t < \alpha - h \\ \Downarrow \\ \alpha - h \text{ is an upper bound of } E \\ \Downarrow \\ \perp\end{gathered}$$

Therefore,

$$\alpha^n = x$$

This guarantees the *existence* of y. Let $0 < \alpha < \beta < \infty$ be given. Then

$$\begin{gathered}\beta^n - \alpha^n = (\beta-\alpha)\sum_{k=0}^{n-1}\beta^{n-1-k}\alpha^k > 0 \\ \Downarrow \\ \beta^n > \alpha^n\end{gathered}$$

This guarantees the *uniqueness* of y. □

Definition 1.6.7. Let $y \in [0,1)$ and $b \in \mathbb{N}_2$ be given. A sequence $(a_n)_{n=1}^{\infty}$ is called a **greedy base b expansion** of y if

$$\forall n \in \mathbb{N}, 0 \le y - \sum_{k=1}^{n}\frac{a_k}{b^k} < \frac{1}{b^n}$$

where

$$\forall k \in \mathbb{N}, a_k \in \mathbb{N}_0^{b-1}$$

In particular, if $b = 10$, such a sequence is called a **greedy decimal expansion** of y.

Theorem 1.6.8. *Let $y \in [0, 1)$ and $b \in \mathbb{N}_2$ be given. Then the greedy base b expansion of y exists and is unique.*

Proof. Define

$$\begin{array}{ccc} t_1 := by & T_1 := \left\{ t \in \mathbb{N}_0^{b-1} \,\middle|\, t \leq t_1 \right\} & a_1 := \max T_1 \\ t_2 := b^2 y - ba_1 & T_2 := \left\{ t \in \mathbb{N}_0^{b-1} \,\middle|\, t \leq t_2 \right\} & a_2 := \max T_2 \\ \vdots & \vdots & \vdots \\ t_n := b^n y - \displaystyle\sum_{k=1}^{n-1} a_k b^{n-k} & T_n := \left\{ t \in \mathbb{N}_0^{b-1} \,\middle|\, t \leq t_n \right\} & a_n := \max T_n \end{array}$$

Let $m \in \mathbb{N}$ be given. If $0 \leq t_m < b$, then

- $0 \in T_m$
- T_m is bounded above by $b - 1$

By Theorem 1.4.4,

$$\begin{gathered} \exists M \in \mathbb{N}_0^{b-1}, M = \max T_m \\ \Downarrow \\ a_m = M \text{ is well-defined} \end{gathered}$$

Define $P(n)$:

- $0 \leq t_n < b$
- $\forall k \in \mathbb{N}^n, a_k = \max T_k$ is well-defined

We will show that $P(n)$ is true for all $n \in \mathbb{N}$.

- **Base case:** Since $0 \leq t_1 = by < b$,
  - $0 \in T_1$
  - T_1 is bounded above by $b - 1$

  By Theorem 1.4.4,

$$\begin{gathered} a_1 = \max T_1 \text{ is well-defined} \\ \Downarrow \\ P(1) \text{ is true} \end{gathered}$$

- **Inductive step:** Let $P(n)$ be true for $n \in \mathbb{N}$. Then

$$\begin{aligned} t_{n+1} &= b^{n+1}y - \sum_{k=1}^{n} a_k b^{n+1-k} \\ &= b\left(b^n y - \sum_{k=1}^{n} a_k b^{n-k}\right) \\ &= b\left(b^n y - \sum_{k=1}^{n-1} a_k b^{n-k} - a_n\right) \\ &= b(t_n - a_n) \end{aligned}$$

  By the inductive hypothesis,

  - $0 \leq t_n < b$
  - $a_n = \max T_n$

  Then

$$\begin{gathered} a_n \leq t_n < a_n + 1 \\ \Downarrow \\ 0 \leq t_n - a_n < 1 \\ \Downarrow \\ 0 \leq t_{n+1} = b(t_n - a_n) < b \\ \Downarrow \\ P(n+1) \text{ is true} \end{gathered}$$

By the *principle of mathematical induction*, $P(n)$ is true for all $n \in \mathbb{N}$. Then

$$\begin{gathered} a_n \leq b^n y - \sum_{k=1}^{n-1} a_k b^{n-k} < a_n + 1 \\ \Downarrow \\ 0 \leq b^n y - \sum_{k=1}^{n} a_k b^{n-k} < 1 \\ \Downarrow \\ 0 \leq y - \sum_{k=1}^{n} \frac{a_k}{b^k} < \frac{1}{b^n} \end{gathered}$$

for all $n \in \mathbb{N}$. This guarantees the *existence* of a *greedy base b expansion* of y. Let two sequences $(a_n)_{n=1}^{\infty}$ and $(c_n)_{n=1}^{\infty}$ satisfy

- $\forall n \in \mathbb{N}, 0 \leq y - \sum_{k=1}^{n} \frac{a_k}{b^k} < \frac{1}{b^n}$
- $\forall n \in \mathbb{N}, 0 \leq y - \sum_{k=1}^{n} \frac{c_k}{b^k} < \frac{1}{b^n}$

where

$$\forall k \in \mathbb{N}, a_k \in \mathbb{N}_0^{b-1} \wedge c_k \in \mathbb{N}_0^{b-1}$$

We will show that

$$\forall n \in \mathbb{N}, a_n = c_n$$

Define

$$Q(n) : \forall k \in \mathbb{N}^n, a_k = c_k$$

We will show that $Q(n)$ is true for all $n \in \mathbb{N}$.

- **Base case:** By the assumption,

$$\begin{array}{ccc} 0 \le y - \dfrac{a_1}{b} < \dfrac{1}{b} & \qquad & 0 \le y - \dfrac{c_1}{b} < \dfrac{1}{b} \\ \Downarrow & & \Downarrow \\ \dfrac{a_1}{b} \le y < \dfrac{a_1+1}{b} & & \dfrac{c_1}{b} \le y < \dfrac{c_1+1}{b} \end{array}$$

  By combining the two inequalities,

$$\begin{array}{ccc} \dfrac{a_1}{b} \le y < \dfrac{c_1+1}{b} & \qquad & \dfrac{c_1}{b} \le y < \dfrac{a_1+1}{b} \\ \Downarrow & & \Downarrow \\ a_1 < c_1 + 1 & & c_1 < a_1 + 1 \\ \Downarrow & & \Downarrow \\ a_1 \le c_1 & & c_1 \le a_1 \end{array}$$

  Therefore,

$$\begin{array}{c} a_1 = c_1 \\ \Downarrow \\ Q(1) \text{ is true} \end{array}$$

- **Inductive step:** Let $Q(n)$ be true for $n \in \mathbb{N}$. Then

$$\begin{array}{ccc} 0 \le y - \displaystyle\sum_{k=1}^{n+1} \frac{a_k}{b^k} < \frac{1}{b^{n+1}} & \qquad & 0 \le y - \displaystyle\sum_{k=1}^{n+1} \frac{c_k}{b^k} < \frac{1}{b^{n+1}} \\ \Downarrow & & \Downarrow \\ \displaystyle\sum_{k=1}^{n+1} \frac{a_k}{b^k} \le y < \sum_{k=1}^{n+1} \frac{a_k}{b^k} + \frac{1}{b^{n+1}} & & \displaystyle\sum_{k=1}^{n+1} \frac{c_k}{b^k} \le y < \sum_{k=1}^{n+1} \frac{c_k}{b^k} + \frac{1}{b^{n+1}} \end{array}$$

  By combining the two inequalities,

$$\begin{array}{ccc} \displaystyle\sum_{k=1}^{n+1} \frac{a_k}{b^k} \le y < \sum_{k=1}^{n+1} \frac{c_k}{b^k} + \frac{1}{b^{n+1}} & \qquad & \displaystyle\sum_{k=1}^{n+1} \frac{c_k}{b^k} \le y < \sum_{k=1}^{n+1} \frac{a_k}{b^k} + \frac{1}{b^{n+1}} \\ \Downarrow & & \Downarrow \\ \displaystyle\sum_{k=1}^{n+1} \frac{a_k - c_k}{b^k} < \frac{1}{b^{n+1}} & & \displaystyle\sum_{k=1}^{n+1} \frac{c_k - a_k}{b^k} < \frac{1}{b^{n+1}} \end{array}$$

By the inductive hypothesis,

$$\forall k \in \mathbb{N}^n, a_k = c_k$$

Then

$$\begin{array}{ccc}
\dfrac{a_{n+1} - c_{n+1}}{b^{n+1}} < \dfrac{1}{b^{n+1}} & & \dfrac{c_{n+1} - a_{n+1}}{b^{n+1}} < \dfrac{1}{b^{n+1}} \\
\Downarrow & & \Downarrow \\
a_{n+1} - c_{n+1} < 1 & & c_{n+1} - a_{n+1} < 1 \\
\Downarrow & & \Downarrow \\
a_{n+1} \leq c_{n+1} & & c_{n+1} \leq a_{n+1}
\end{array}$$

Therefore,

$$\begin{gathered}
a_{n+1} = c_{n+1} \\
\Downarrow \\
Q(n+1) \text{ is true}
\end{gathered}$$

By the *principle of mathematical induction*, $Q(n)$ is true for all $n \in \mathbb{N}$. This guarantees the *uniqueness* of the *greedy base b expansion* of y. □

Theorem 1.6.9. *Let $b \in \mathbb{N}_2$ be given. Let*

- $(a_n)_{n=1}^{\infty}$ *be the greedy base b expansion of $a \in [0, 1)$*
- $(c_n)_{n=1}^{\infty}$ *be the greedy base b expansion of $c \in [0, 1)$*

Then

$$(\forall n \in \mathbb{N}, a_n = c_n) \implies a = c$$

Proof. By Corollary 1.6.2, if $a \neq c$ (*WLOG* $a < c$), then

$$\exists n \in \mathbb{N}, \frac{1}{b^n} < c - a$$

By Definition 1.6.7,

$$\begin{array}{ccc}
0 \leq a - \displaystyle\sum_{k=1}^{n} \frac{a_k}{b^k} < \frac{1}{b^n} & & 0 \leq c - \displaystyle\sum_{k=1}^{n} \frac{c_k}{b^k} < \frac{1}{b^n} \\
\Downarrow & & \Downarrow \\
\displaystyle\sum_{k=1}^{n} \frac{a_k}{b^k} + \frac{1}{b^n} \leq a + \frac{1}{b^n} & & c < \displaystyle\sum_{k=1}^{n} \frac{c_k}{b^k} + \frac{1}{b^n}
\end{array}$$

By the assumption,

$$\begin{gathered}\forall n \in \mathbb{N}, a_n = c_n \\ \Downarrow \\ \sum_{k=1}^{n} \frac{a_k}{b^k} = \sum_{k=1}^{n} \frac{c_k}{b^k} \\ \Downarrow \\ \sum_{k=1}^{n} \frac{a_k}{b^k} + \frac{1}{b^n} \leq a + \frac{1}{b^n} < c < \sum_{k=1}^{n} \frac{c_k}{b^k} + \frac{1}{b^n} \\ \Downarrow \\ \perp\end{gathered}$$

Therefore,

$$a = c$$

□

Theorem 1.6.10. *Let $b \in \mathbb{N}_2$ be given. Let $(a_n)_{n=1}^{\infty}$ be such that*

$$\forall k \in \mathbb{N}, a_k \in \mathbb{N}_0^{b-1}$$

If

$$\nexists y \in [0,1), (a_n)_{n=1}^{\infty} \textit{ is the greedy base } b \textit{ expansion of } y$$

then

$$\exists N \in \mathbb{N}, \forall n \in \mathbb{N}_N, a_n = b - 1$$

Proof. We will prove the contrapositive: If

$$\forall N \in \mathbb{N}, \exists n \in \mathbb{N}_N, a_n \neq b - 1$$

then

$$\exists y \in [0,1), (a_n)_{n=1}^{\infty} \text{ is the } \textit{greedy base } b \textit{ expansion} \text{ of } y$$

Assume that

$$\forall N \in \mathbb{N}, \exists n \in \mathbb{N}_N, a_n \neq b - 1$$

Define

$$x_n := \sum_{k=1}^{n} \frac{a_k}{b^k}$$

for each $n \in \mathbb{N}$. Then

$$\begin{gathered}\forall k \in \mathbb{N}, a_k \in \mathbb{N}_0^{b-1} \\ \Downarrow \\ x_n = \sum_{k=1}^{n} \frac{a_k}{b^k} \leq \sum_{k=1}^{n} \frac{b-1}{b^k} = 1 - \frac{1}{b^n} < 1\end{gathered}$$

for all $n \in \mathbb{N}$. Therefore, $\{x_n\}_{n=1}^{\infty}$ is bounded above by 1. By the *least upper bound property*,

$$\exists y \in \mathbb{R}, y = \sup\{x_n\}_{n=1}^{\infty}$$

First, we will show that $y \in [0, 1)$. By the assumption,

$$\exists m \in \mathbb{N}, a_m \neq b - 1$$

Then

$$x_n = x_m + \sum_{k=m+1}^{n} \frac{a_k}{b^k} \leq x_m + \sum_{k=m+1}^{n} \frac{b-1}{b^k}$$
$$\Downarrow$$
$$x_n \leq x_m + \frac{1}{b^m} - \frac{1}{b^n} < x_m + \frac{1}{b^m}$$

for all $n > m$. Since $a_m \leq b - 2$,

$$\begin{aligned} x_m + \frac{1}{b^m} &= \sum_{k=1}^{m-1} \frac{a_k}{b^k} + \frac{a_m + 1}{b^m} \\ &\leq \sum_{k=1}^{m-1} \frac{a_k}{b^k} + \frac{b-1}{b^m} \\ &\leq \sum_{k=1}^{m} \frac{b-1}{b^k} = 1 - \frac{1}{b^m} \end{aligned}$$

To sum up,

- $\forall n \in \mathbb{N}, n \leq m \implies x_n \leq x_m$
- $\forall n \in \mathbb{N}, n > m \implies x_n < x_m + \dfrac{1}{b^m} \leq 1 - \dfrac{1}{b^m}$

Therefore, $1 - \dfrac{1}{b^m}$ is an upper bound of $\{x_n\}_{n=1}^{\infty}$. Since $0 \leq x_1 \leq y$,

$$y = \sup\{x_n\}_{n=1}^{\infty}$$
$$\Downarrow$$
$$y \leq 1 - \frac{1}{b^m} < 1$$
$$\Downarrow$$
$$y \in [0, 1)$$

Next, we will show that $(a_n)_{n=1}^{\infty}$ is the *greedy base b expansion* of y. Let $n \in \mathbb{N}$ be fixed. By the assumption,

$$\exists m \in \mathbb{N}_{n+1}, a_m \leq b - 2$$

Then

$$x_m = x_n + \sum_{k=n+1}^{m} \frac{a_k}{b^k} < x_n + \sum_{k=n+1}^{m} \frac{b-1}{b^k}$$
$$\Downarrow$$
$$x_m < x_n + \frac{1}{b^n} - \frac{1}{b^m}$$
$$\Downarrow$$
$$x_m + \frac{1}{b^m} < x_n + \frac{1}{b^n}$$

By the same argument as above, $x_m + \frac{1}{b^m}$ is an upper bound of $\{x_n\}_{n=1}^{\infty}$. Then

$$y = \sup\{x_n\}_{n=1}^{\infty}$$
$$\Downarrow$$
$$y \leq x_m + \frac{1}{b^m} < x_n + \frac{1}{b^n}$$
$$\Downarrow$$
$$0 \leq y - x_n < \frac{1}{b^n}$$

Since n was arbitrary, $(a_n)_{n=1}^{\infty}$ is the *greedy base* b *expansion* of y. □

Theorem 1.6.11. *Let* $b \in \mathbb{N}_2$ *be given. Let* $(a_n)_{n=1}^{\infty}$ *be such that*

$$\forall k \in \mathbb{N}, a_k \in \mathbb{N}_0^{b-1}$$

If

$$\exists N \in \mathbb{N}, \forall n \in \mathbb{N}_N, a_n = b - 1$$

then

$$\nexists y \in [0, 1), (a_n)_{n=1}^{\infty} \textit{ is the greedy base } b \textit{ expansion of } y$$

This theorem is the ***converse*** *of Theorem 1.6.10.*

Proof. Assume that

$$\exists N \in \mathbb{N}, \forall n \in \mathbb{N}_N, a_n = b - 1$$

If

$$\exists y \in [0, 1), (a_n)_{n=1}^{\infty} \text{ is the } \textit{greedy base } b \textit{ expansion} \text{ of } y$$

then

$$0 \leq y - \sum_{k=1}^{n} \frac{a_k}{b^k} < \frac{1}{b^n}$$

for all $n \in \mathbb{N}$. Define

$$x_n := \sum_{k=1}^{n} \frac{a_k}{b^k}$$

for each $n \in \mathbb{N}$. Then

$$x_n = x_N + \sum_{k=N+1}^{n} \frac{b-1}{b^k} = x_N + \frac{1}{b^N} - \frac{1}{b^n}$$
$$\Downarrow$$
$$y - x_n = y - x_N - \frac{1}{b^N} + \frac{1}{b^n} \geq 0$$
$$\Downarrow$$
$$\frac{1}{b^n} \geq x_N + \frac{1}{b^N} - y$$

for all $n > N$. Since $N \in \mathbb{N}$,

$$0 \leq y - x_N < \frac{1}{b^N}$$
$$\Downarrow$$
$$x_N + \frac{1}{b^N} - y > 0$$

By Corollary 1.6.2,

$$\exists m \in \mathbb{N}, \frac{1}{b^m} < x_N + \frac{1}{b^N} - y$$
$$\Downarrow$$
$$\forall n \in \mathbb{N}, n > \max\{m, N\} \implies \frac{1}{b^n} < x_N + \frac{1}{b^N} - y$$
$$\Downarrow$$
$$\perp$$

Therefore,

$$\nexists y \in [0,1), (a_n)_{n=1}^{\infty} \text{ is the } \textit{greedy base } b \textit{ expansion} \text{ of } y$$

□

1.7 The Extended Real Number System

The Extended real number system $\overline{\mathbb{R}}$ is defined as

$$\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, +\infty\}$$

We define the following operations on $\overline{\mathbb{R}}$:

- $\forall x \in \mathbb{R} \cup \{+\infty\}, x + (+\infty) := +\infty$
- $\forall x \in \mathbb{R} \cup \{-\infty\}, x + (-\infty) := -\infty$
- $\forall x \in \mathbb{R}^{+} \cup \{+\infty\}, x \cdot (\pm\infty) := \pm\infty$
- $\forall x \in \mathbb{R}^{-} \cup \{-\infty\}, x \cdot (\pm\infty) := \mp\infty$

- $\forall x \in \mathbb{R}, \dfrac{x}{\pm\infty} := 0$

The following operations are undefined in $\overline{\mathbb{R}}$:

- $(\pm\infty) + (\mp\infty)$
- $0 \cdot (\pm\infty)$
- $\dfrac{(\pm\infty)}{(\pm\infty)}$ and $\dfrac{(\pm\infty)}{(\mp\infty)}$
- $\dfrac{x}{0}$ for all $x \in \overline{\mathbb{R}}$

$\overline{\mathbb{R}}$ is not a field, but it is an ordered set with the following order relation:

$$\forall x \in \mathbb{R}, -\infty < x < +\infty$$

Remark. We can identify the following intervals in $\overline{\mathbb{R}}$:

- $\forall a \in \mathbb{R}, (-\infty, a) = \{x \in \mathbb{R} \mid x < a\}$
- $\forall a \in \mathbb{R}, (a, +\infty) = \{x \in \mathbb{R} \mid x > a\}$
- $(-\infty, +\infty) = \mathbb{R}$.

Theorem 1.7.1. *For all $E \subseteq \overline{\mathbb{R}}$,*

- $\exists x \in \overline{\mathbb{R}}, x = \sup E$
- $\exists y \in \overline{\mathbb{R}}, y = \inf E$

Proof. If $E = \varnothing$, then

- $\sup E = -\infty$
- $\inf E = +\infty$

Otherwise,

- If E is bounded above in $\mathbb{R}$, then

  $$\exists x \in \mathbb{R}, x = \sup E$$

  by the *least upper bound property*.

- Otherwise, $\sup E = +\infty$

Also,

- If E is bounded below in $\mathbb{R}$, then

  $$\exists y \in \mathbb{R}, y = \inf E$$

  by Theorem 1.4.8.

- Otherwise, $\inf E = -\infty$

□

1.8 The Complex Number System

The complex number system $\mathbb{C}$ is defined as

$$\mathbb{C} \coloneqq \{(x, y) \mid x, y \in \mathbb{R}\}$$

The binary operations on $\mathbb{C}$ are defined as

- $\forall x, y, u, v \in \mathbb{R}, (x, y) + (u, v) \coloneqq (x + u, y + v)$
- $\forall x, y, u, v \in \mathbb{R}, (x, y) \cdot (u, v) \coloneqq (xu - yv, xv + yu)$

One can verify that $\mathbb{C}$ is a field with the following properties:

- The **additive identity** is $(0, 0)$
- The **multiplicative identity** is $(1, 0)$
- The **additive inverse** of (x, y) is $(-x, -y)$
- The **multiplicative inverse** of (x, y) is
$$\left(\frac{x}{x^2 + y^2}, -\frac{y}{x^2 + y^2}\right)$$
for $(x, y) \neq (0, 0)$

The **imaginary unit** i is defined as

$$i \coloneqq (0, 1) \in \mathbb{C}$$

Then
$$i^2 = (0, 1) \cdot (0, 1) = (-1, 0)$$

There is a **canonical embedding**

$$\mathbb{R} \xrightarrow{\cong} \mathbb{C} \text{ such that } x \mapsto (x, 0)$$

Therefore, we identify $\mathbb{R}$ with its image in $\mathbb{C}$ and write

$$\mathbb{R} \subseteq \mathbb{C}$$

Then, for every complex number $(x, y) \in \mathbb{C}$,

$$(x, y) = (x, 0) + (0, y) = (x, 0) + (y, 0) \cdot (0, 1) = x + yi$$

Definition 1.8.1. Let $z = (x, y) \in \mathbb{C}$ be given.

- The **real part** of z is defined as

  - $\Re(z) := x$
  - $\mathrm{Re}(z) := x$
- The **imaginary part** of z is defined as
  - $\Im(z) := y$
  - $\mathrm{Im}(z) := y$
- The **complex conjugate** of z is defined as

$$\overline{z} := (x, -y) = x - yi$$

Theorem 1.8.2. *Let $z = (x, y) \in \mathbb{C}$ and $w = (u, v) \in \mathbb{C}$ be given. Then the following properties hold:*

- $\overline{(\overline{z})} = z$
- $z = \overline{z} \iff z \in \mathbb{R}$
- $\overline{z + w} = \overline{z} + \overline{w}$
- $\overline{zw} = \overline{z} \cdot \overline{w}$
- $z + \overline{z} = 2\,\mathrm{Re}(z)$
- $z - \overline{z} = 2i\,\mathrm{Im}(z)$
- $z\overline{z} = x^2 + y^2 \in \mathbb{R}^+ \cup \{0\}$

Proof. The proof follows directly from Definition 1.8.1. □

Definition 1.8.3. Let $z = (x, y) \in \mathbb{C}$ be given. The **modulus** of z is defined as

$$|z| := \sqrt{z\overline{z}} = \sqrt{x^2 + y^2} \in \mathbb{R}^+ \cup \{0\}$$

The *existence* and *uniqueness* of $|z|$ are guaranteed by Theorems 1.6.6 and 1.8.2

Theorem 1.8.4. *Let $a_1, a_2, \ldots, a_n \in \mathbb{C}$ and $b_1, b_2, \ldots, b_n \in \mathbb{C}$ be given. Then*

$$\left|\sum_{k=1}^{n} a_k \overline{b_k}\right|^2 \leq \sum_{k=1}^{n} |a_k|^2 \cdot \sum_{k=1}^{n} |b_k|^2$$

*This theorem is called the **Cauchy-Schwarz inequality**.*

Proof. Let $1 \le i < j \le n$ be given. Then

$$\begin{aligned} |a_i b_j - a_j b_i|^2 &= (a_i b_j - a_j b_i)\overline{(a_i b_j - a_j b_i)} \\ &= |a_i|^2|b_j|^2 + |a_j|^2|b_i|^2 - (a_i\overline{b_i} \cdot \overline{a_j} b_j + \overline{a_i} b_i \cdot a_j \overline{b_j}) \end{aligned}$$

Consider the following sums:

$$\begin{aligned} \sum_{i<j}(|a_i|^2|b_j|^2 + |a_j|^2|b_i|^2) &= \sum_{i<j}(|a_i|^2|b_j|^2) + \sum_{j<i}(|a_i|^2|b_j|^2) \\ &= \sum_{i \neq j}(|a_i|^2|b_j|^2) \\ &= \sum_{k=1}^{n} |a_k|^2 \cdot \sum_{k=1}^{n} |b_k|^2 - \sum_{k=1}^{n} |a_k|^2|b_k|^2 \end{aligned}$$

$$\begin{aligned} \sum_{i<j}(a_i\overline{b_i} \cdot \overline{a_j} b_j + \overline{a_i} b_i \cdot a_j \overline{b_j}) &= \sum_{i \neq j}(a_i\overline{b_i} \cdot \overline{a_j} b_j) \\ &= \sum_{i=1}^{n} a_i\overline{b_i} \cdot \sum_{j=1}^{n} \overline{a_j} b_j - \sum_{k=1}^{n} a_k\overline{b_k} \cdot \overline{a_k} b_k \\ &= \left|\sum_{k=1}^{n} a_k\overline{b_k}\right|^2 - \sum_{k=1}^{n} |a_k|^2|b_k|^2 \end{aligned}$$

Therefore,

$$\sum_{i<j}|a_i b_j - a_j b_i|^2 = \sum_{k=1}^{n} |a_k|^2 \cdot \sum_{k=1}^{n} |b_k|^2 - \left|\sum_{k=1}^{n} a_k\overline{b_k}\right|^2 \ge 0$$

$\square$

Corollary 1.8.5. *Let $a_1, a_2, \ldots, a_n \in \mathbb{R}$ and $b_1, b_2, \ldots, b_n \in \mathbb{R}$ be given. Then*

$$\left(\sum_{k=1}^{n} a_k b_k\right)^2 \le \sum_{k=1}^{n} a_k^2 \cdot \sum_{k=1}^{n} b_k^2$$

Theorem 1.8.6. *Let $z, w \in \mathbb{C}$ be given. Then the following properties hold:*

(A) $|zw| = |z| \cdot |w|$

(B) $|z + w| \le |z| + |w|$

(C) $||z| - |w|| \le |z - w|$

Proof. Let $z = (x, y) \in \mathbb{C}$ and $w = (u, v) \in \mathbb{C}$ be given.

(A) $|zw|^2 = (xu - yv)^2 + (xv + yu)^2 = (x^2 + y^2)(u^2 + v^2) = |z|^2 \cdot |w|^2$

(B) By Corollary 1.8.5,

$$(xu + yv)^2 \leq (x^2 + y^2)(u^2 + v^2)$$

Then

$$\begin{aligned} |z + w|^2 &= (x + u)^2 + (y + v)^2 \\ &= x^2 + y^2 + u^2 + v^2 + 2(xu + yv) \\ &\leq x^2 + y^2 + u^2 + v^2 + 2\sqrt{x^2 + y^2} \cdot \sqrt{u^2 + v^2} \\ &= (|z| + |w|)^2 \end{aligned}$$

(C) Again, by Corollary 1.8.5,

$$\begin{aligned} |z - w|^2 &= (x - u)^2 + (y - v)^2 \\ &= x^2 + y^2 + u^2 + v^2 - 2(xu + yv) \\ &\geq x^2 + y^2 + u^2 + v^2 - 2\sqrt{x^2 + y^2} \cdot \sqrt{u^2 + v^2} \\ &= (|z| - |w|)^2 \end{aligned}$$

□

1.9 Euclidean Spaces

For each $n \in \mathbb{N}$, the **Euclidean n-space** $\mathbb{R}^n$ is defined as

$$\mathbb{R}^n := \{\boldsymbol{x} = (x_1, x_2, \ldots, x_n) \mid \forall k \in \mathbb{N}^n, x_k \in \mathbb{R}\}$$

where $x_1, x_2, \ldots, x_n$ are called the **coordinates** of $\boldsymbol{x}$. The elements of $\mathbb{R}^n$ for $n > 1$ are called **points** (or *vectors*) and are represented by bold symbols.

The operations on $\mathbb{R}^n$ are defined as

- $\forall \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n, \boldsymbol{x} + \boldsymbol{y} := (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n)$
- $\forall \alpha \in \mathbb{R}, \forall \boldsymbol{x} \in \mathbb{R}^n, \alpha\boldsymbol{x} := (\alpha x_1, \alpha x_2, \ldots, \alpha x_n)$

Definition 1.9.1. Let $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ be given.

- The **inner product** of $\boldsymbol{x}$ and $\boldsymbol{y}$ is defined as

$$\boldsymbol{x} \cdot \boldsymbol{y} := \sum_{k=1}^{n} x_k y_k$$

- The **norm** of $\boldsymbol{x}$ is defined as

$$\|\boldsymbol{x}\| := \sqrt{\boldsymbol{x} \cdot \boldsymbol{x}} = \sqrt{\sum_{k=1}^{n} x_k^2}$$

- The **distance** between $\boldsymbol{x}$ and $\boldsymbol{y}$ is defined as

$$d(\boldsymbol{x}, \boldsymbol{y}) := \|\boldsymbol{x} - \boldsymbol{y}\| = \sqrt{\sum_{k=1}^{n} (x_k - y_k)^2}$$

where one can verify that $d(x, y) = |x - y|$ in the case of $n = 1$.

Theorem 1.9.2. *Let $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathbb{R}^n$ be given. Then the following properties hold:*

(A) $|\boldsymbol{x} \cdot \boldsymbol{y}| \leq \|\boldsymbol{x}\| \cdot \|\boldsymbol{y}\|$

(B) $\|\boldsymbol{x} + \boldsymbol{y}\| \leq \|\boldsymbol{x}\| + \|\boldsymbol{y}\|$

(C) $\|\boldsymbol{x} - \boldsymbol{y}\| \geq |\|\boldsymbol{x}\| - \|\boldsymbol{y}\||$

Proof. **(A)** By Corollary 1.8.5,

$$|\boldsymbol{x} \cdot \boldsymbol{y}|^2 = \left|\sum_{k=1}^{n} x_k y_k\right|^2 \leq \sum_{k=1}^{n} x_k^2 \cdot \sum_{k=1}^{n} y_k^2 = \|\boldsymbol{x}\|^2 \cdot \|\boldsymbol{y}\|^2$$

(B) By part **(A)**,

$$\begin{aligned} \|\boldsymbol{x} + \boldsymbol{y}\|^2 &= (\boldsymbol{x} + \boldsymbol{y}) \cdot (\boldsymbol{x} + \boldsymbol{y}) = \|\boldsymbol{x}\|^2 + 2\boldsymbol{x} \cdot \boldsymbol{y} + \|\boldsymbol{y}\|^2 \\ &\leq \|\boldsymbol{x}\|^2 + 2|\boldsymbol{x} \cdot \boldsymbol{y}| + \|\boldsymbol{y}\|^2 \leq \|\boldsymbol{x}\|^2 + 2\|\boldsymbol{x}\| \cdot \|\boldsymbol{y}\| + \|\boldsymbol{y}\|^2 \\ &= (\|\boldsymbol{x}\| + \|\boldsymbol{y}\|)^2 \end{aligned}$$

(C) By part **(B)**,

$$\begin{gathered} \|\boldsymbol{x}\| = \|(\boldsymbol{x} - \boldsymbol{y}) + \boldsymbol{y}\| \leq \|\boldsymbol{x} - \boldsymbol{y}\| + \|\boldsymbol{y}\| \\ \Downarrow \\ \|\boldsymbol{x}\| - \|\boldsymbol{y}\| \leq \|\boldsymbol{x} - \boldsymbol{y}\| \end{gathered}$$

Also,

$$\begin{gathered} \|\boldsymbol{y}\| = \|(\boldsymbol{y} - \boldsymbol{x}) + \boldsymbol{x}\| \leq \|\boldsymbol{y} - \boldsymbol{x}\| + \|\boldsymbol{x}\| \\ \Downarrow \\ \|\boldsymbol{y}\| - \|\boldsymbol{x}\| \leq \|\boldsymbol{x} - \boldsymbol{y}\| \end{gathered}$$

To sum up,

$$\|\boldsymbol{x} - \boldsymbol{y}\| \geq |\|\boldsymbol{x}\| - \|\boldsymbol{y}\||$$

$\square$

Chapter 2

Basic Topology

2.1 Functions

A **function** (or *mapping*) is defined with the following three components:

- A **domain** X
- A **codomain** Y
- A **rule** f which assigns a *unique* element $y \in Y$ to each $x \in X$

f is said to be a function from X to Y, which is denoted by

$$f : X \to Y$$

For each $x \in X$, the *unique* element $y \in Y$ assigned to x is denoted by

$$y = f(x)$$

Definition 2.1.1. Let $f : X \to Y$ be given. For every $E \subseteq X$, the **image** of E under f is defined as

$$f(E) \coloneqq \{f(x) \mid x \in E\} \subseteq Y$$

In particular, if $E = X$, then $f(X) \subseteq Y$ is called the **range** of f.

Definition 2.1.2. Let $f : X \to Y$ be given. For every $E \subseteq Y$, the **inverse image** of E under f is defined as

$$f^{-1}(E) \coloneqq \{x \in X \mid f(x) \in E\} \subseteq X$$

Definition 2.1.3. Let $f : X \to Y$ be given.

- f is **injective** (or *one-to-one*) if

$$\forall x_1, x_2 \in X, f(x_1) = f(x_2) \implies x_1 = x_2$$

- f is **surjective** (or *onto*) if

$$\forall y \in Y, \exists x \in X, f(x) = y$$

  or equivalently,

$$f(X) = Y$$

- f is **bijective** if it is both injective and surjective

Remark. The following notations are used to denote the properties of functions:

- $f : X \hookrightarrow Y$ denotes that f is injective
- $f : X \twoheadrightarrow Y$ denotes that f is surjective
- $f : X \xrightarrow{\sim} Y$ or $f : X \xrightarrow{\cong} Y$ denotes that f is bijective

Definition 2.1.4. Let $f : X \xrightarrow{\sim} Y$ be given. The **inverse function** $f^{-1} : Y \to X$ of f is defined as

$$\forall x \in X, f^{-1}(f(x)) = x$$

If

- $f^{-1}(y) = x_1$
- $f^{-1}(y) = x_2$

then

- $f(x_1) = y$
- $f(x_2) = y$

Since f is injective,

$$\begin{gathered} f(x_1) = f(x_2) \\ \Downarrow \\ x_1 = x_2 \end{gathered}$$

Since f is surjective,

$$\forall y \in Y, \exists x \in X, f(x) = y$$

Therefore, f^{-1} is well-defined.

Theorem 2.1.5. *Let $f : X \xrightarrow{\sim} Y$ be given. Then $f^{-1} : Y \to X$ is bijective.*

Proof. Let $x \in X$ and $y_1, y_2 \in Y$ be given.

- **Injectivity:** If
$$f^{-1}(y_1) = f^{-1}(y_2) = x$$
then, by Definition 2.1.4,
  - $f(x) = y_1$
  - $f(x) = y_2$

  Since f is a function,
$$y_1 = y_2$$
- **Surjectivity:** Since f is a function,
$$\begin{gathered} \forall x \in X, \exists y \in Y, f(x) = y \\ \Downarrow \\ f^{-1}(y) = x \\ \Downarrow \\ \forall x \in X, \exists y \in Y, f^{-1}(y) = x \end{gathered}$$

$\square$

Definition 2.1.6. Let $E \subseteq X$, $F \subseteq Y$, and $f : X \to Y$ be given. The following **restriction functions** are defined as

- The **restriction** $f|_E : E \to Y$ of f to E is defined as
$$\forall x \in E, f|_E(x) = f(x)$$
- The **corestriction** $f|^F : X \to F$ of f to F is defined as
$$\forall x \in X, f|^F(x) = f(x)$$
for $f(X) \subseteq F$
- The **bi-restriction** $f|_E^F : E \to F$ of f to E and F is defined as
$$\forall x \in E, f|_E^F(x) = f(x)$$
for $f(E) \subseteq F$

Theorem 2.1.7. *Let $f : X \to Y$ be given. Let $E \subseteq X$ and $F \subseteq Y$ be such that*

$$f(E) \subseteq F$$

Then the following properties hold:

(A) *f is injective $\implies$ $f|_E^F$ is injective*

(B) *$f(E) = F \implies f|_E^F$ is surjective*

(C) *If*

- *f is injective*
- *$f(E) = F$*

then $f|_E^F$ is bijective

Proof. Let $x_1, x_2 \in E$ be given.

(A) If f is injective, then

$$\begin{gathered} f|_E^F(x_1) = f|_E^F(x_2) \\ \Downarrow \\ f(x_1) = f(x_2) \\ \Downarrow \\ x_1 = x_2 \end{gathered}$$

(B) If $f(E) = F$, then

$$\begin{gathered} \forall y \in F, \exists x \in E, f(x) = y \\ \Downarrow \\ \forall y \in F, \exists x \in E, f|_E^F(x) = y \end{gathered}$$

(C) follows from **(A)** and **(B)**

□

Definition 2.1.8. Let $f : X \to Y$ and $g : Y \to Z$ be given. The **composition** $g \circ f : X \to Z$ of g with f is defined as

$$\forall x \in X, (g \circ f)(x) = g(f(x))$$

Theorem 2.1.9. *Let $f : X \to Y$ and $g : Y \to Z$ be given. Then the following properties hold:*

(A) *f and g are injective $\implies$ $g \circ f$ is injective*

(B) *f and g are surjective $\implies$ $g \circ f$ is surjective*

(C) *f and g are bijective $\implies$ $g \circ f$ is bijective*

Proof. Let $x_1, x_2 \in X$ be given.

(A) If f and g are injective, then

$$\begin{gathered}(g \circ f)(x_1) = (g \circ f)(x_2) \\ \Downarrow \\ g(f(x_1)) = g(f(x_2)) \\ \Downarrow \\ f(x_1) = f(x_2) \\ \Downarrow \\ x_1 = x_2\end{gathered}$$

(B) If f and g are surjective, then

$$\forall z \in Z, \exists y \in Y, g(y) = z$$

Also,

$$\forall y \in Y, \exists x \in X, f(x) = y$$

Therefore,

$$\forall z \in Z, \exists x \in X, (g \circ f)(x) = z$$

(C) follows from **(A)** and **(B)**

□

Theorem 2.1.10. *Let S be a set of sets. Define*

$$\forall X, Y \in S, X \sim Y \iff \exists f : X \xrightarrow{\sim} Y$$

Then $\sim$ is an equivalence relation on S.

Proof. Let $X, Y, Z \in S$ be given.

- **Reflexivity:**

$$\begin{gathered}\forall X \in S, \exists f : X \xrightarrow{\sim} X, f(x) = x \\ \Downarrow \\ \forall X \in S, X \sim X\end{gathered}$$

- **Symmetry:** If $X \sim Y$, by Theorem 2.1.5,

$$\begin{gathered}\exists f : X \xrightarrow{\sim} Y \\ \Downarrow \\ \exists f^{-1} : Y \xrightarrow{\sim} X\end{gathered}$$

  Therefore,

$$\forall X, Y \in S, X \sim Y \implies Y \sim X$$

- **Transitivity:** If $X \sim Y$ and $Y \sim Z$, by Theorem 2.1.5,

$$\begin{gathered} \exists f : X \xrightarrow{\sim} Y \wedge \exists g : Y \xrightarrow{\sim} Z \\ \Downarrow \\ \exists g \circ f : X \xrightarrow{\sim} Z \end{gathered}$$

Therefore,

$$\forall X, Y, Z \in S, X \sim Y \wedge Y \sim Z \implies X \sim Z$$

□

2.2 Finite Sets

Recall the relation $\sim$ defined in Theorem 2.1.10. A set X is said to be **finite** if

$$\exists n \in \mathbb{N}_0, \mathbb{N}^n \sim X$$

In this case, n is called the **cardinality** of X, which is denoted by

$$|X| = n$$

A set X is said to be

- **infinite** if it is not finite
- **countable** if $\mathbb{N} \sim X$
- **at most countable** if it is either finite or countable
- **uncountable** if it is neither finite nor countable

Remark. In this textbook, we will use the following notations:

- X is finite $\iff |X| < \aleph_0$
- X is infinite $\iff |X| \geq \aleph_0$
- X is countable $\iff |X| = \aleph_0$
- X is at most countable $\iff |X| \leq \aleph_0$
- X is uncountable $\iff |X| > \aleph_0$

The order symbols above are not **total orders**; they are just convenient notations.

Lemma 2.2.1.

$$\forall n \in \mathbb{N}_0, \forall A \subseteq \mathbb{N}^n, |A| < \aleph_0$$

Proof. Define

$$P(n) : \forall A \subseteq \mathbb{N}^n, |A| < \aleph_0$$

We will show that $P(n)$ is true for all $n \in \mathbb{N}_0$.

- **Base case:** Define $f : \mathbb{N}^0 \xrightarrow{\sim} \varnothing$ as

  $$\forall k \in \mathbb{N}^0, f(k) = k$$

  Then

  $$|\varnothing| = 0 < \aleph_0$$

  Therefore, $P(0)$ is true.

- **Inductive step:** Let $P(n)$ be true for $n \in \mathbb{N}_0$. Let $A \subseteq \mathbb{N}^{n+1}$ be given.

  - If $n + 1 \notin A$, then

    $$\begin{gathered} A \subseteq \mathbb{N}^n \\ \Downarrow \\ |A| < \aleph_0 \end{gathered}$$

  - If $n + 1 \in A$, define

    $$B \coloneqq A \setminus \{n + 1\} \subseteq \mathbb{N}^n$$

    Since $B \subseteq \mathbb{N}^n$,

    $$\begin{gathered} B \subseteq \mathbb{N}^n \\ \Downarrow \\ |B| < \aleph_0 \\ \Downarrow \\ \exists m \in \mathbb{N}_0, \exists f : \mathbb{N}^m \xrightarrow{\sim} B \end{gathered}$$

    Define $g : \mathbb{N}^{m+1} \xrightarrow{\sim} A$ as

    $$\forall k \in \mathbb{N}^{m+1}, g(k) = \begin{cases} f(k), & \text{if } k \leq m \\ n + 1, & \text{if } k = m + 1 \end{cases}$$

    Then

    $$|A| = m + 1 < \aleph_0$$

  Therefore, $P(n + 1)$ is true.

By the *principle of mathematical induction*, $P(n)$ is true for all $n \in \mathbb{N}_0$. □

Theorem 2.2.2.

$$\forall X, \forall Y \subseteq X, |X| < \aleph_0 \implies |Y| < \aleph_0$$

Proof. Let X be finite. Then

$$\begin{gathered} |X| < \aleph_0 \\ \Downarrow \\ \exists n \in \mathbb{N}_0, \exists f : \mathbb{N}^n \xrightarrow{\sim} X \end{gathered}$$

Define

$$Z := f^{-1}(Y) \subseteq \mathbb{N}^n$$

for $Y \subseteq X$. By Lemma 2.2.1,

$$\begin{gathered} Z \subseteq \mathbb{N}^n \\ \Downarrow \\ |Z| < \aleph_0 \\ \Downarrow \\ \exists m \in \mathbb{N}_0, \exists g : \mathbb{N}^m \xrightarrow{\sim} Z \end{gathered}$$

By Theorems 2.1.7 and 2.1.9,

$$\begin{gathered} f(Z) = Y \\ \Downarrow \\ \exists f|_Z^Y : Z \xrightarrow{\sim} Y \\ \Downarrow \\ \exists f|_Z^Y \circ g : \mathbb{N}^m \xrightarrow{\sim} Y \\ \Downarrow \\ |Y| < \aleph_0 \end{gathered}$$

$\square$

Lemma 2.2.3.

$$\forall Y, (\exists n \in \mathbb{N}_0, \exists f : \mathbb{N}^n \twoheadrightarrow Y) \implies |Y| < \aleph_0$$

Proof. Assume that

$$\exists n \in \mathbb{N}_0, \exists f : \mathbb{N}^n \twoheadrightarrow Y$$

Define

$$N_y := f^{-1}(\{y\}) \subseteq \mathbb{N}^n$$

for each $y \in Y$. By Theorem 1.1.4,

$$\begin{gathered} f \text{ is surjective} \\ \Downarrow \\ \forall y \in Y, N_y \neq \varnothing \\ \Downarrow \\ \exists n_y \in N_y, \forall x \in N_y, n_y \leq x \end{gathered}$$

Define

$$M_Y := \{n_y \mid y \in Y\} \subseteq \mathbb{N}^n$$

Define $g : M_Y \xrightarrow{\sim} Y$ as

$$\forall n_y \in M_Y, g(n_y) = y$$

By Lemma 2.2.1 and theorem 2.1.9,

$$\begin{gathered} |M_Y| < \aleph_0 \\ \Downarrow \\ \exists m \in \mathbb{N}_0, \exists h : \mathbb{N}^m \xrightarrow{\sim} M_Y \\ \Downarrow \\ \exists g \circ h : \mathbb{N}^m \xrightarrow{\sim} Y \\ \Downarrow \\ |Y| < \aleph_0 \end{gathered}$$

□

Theorem 2.2.4.

$$\forall X \forall Y, (|X| < \aleph_0 \wedge \exists f : X \twoheadrightarrow Y) \implies |Y| < \aleph_0$$

Proof. Assume that

$$|X| < \aleph_0 \wedge \exists f : X \twoheadrightarrow Y$$

By Theorem 2.1.9 and lemma 2.2.3,

$$\begin{gathered} |X| < \aleph_0 \\ \Downarrow \\ \exists n \in \mathbb{N}_0, \exists g : \mathbb{N}^n \xrightarrow{\sim} X \\ \Downarrow \\ \exists f \circ g : \mathbb{N}^n \twoheadrightarrow Y \\ \Downarrow \\ |Y| < \aleph_0 \end{gathered}$$

□

Theorem 2.2.5. *Let X satisfy*

- $0 < |X| < \aleph_0$
- *X is ordered*

Then

- $\exists m \in X, \forall x \in X, m \leq x$

- $\exists M \in X, \forall x \in X, x \leq M$

Proof. Since $0 < |X| < \aleph_0$,

$$\begin{gathered} 0 < |X| < \aleph_0 \\ \Downarrow \\ \exists n \in \mathbb{N}, \exists f : \mathbb{N}^n \xrightarrow{\sim} X \end{gathered}$$

The elements of X can be listed as

$$X = \{a_1, a_2, \ldots, a_n\}$$

where

$$\forall k \in \mathbb{N}, a_k = f(k)$$

Define $P(n)$: If

- $|X| = n$
- X is ordered

then

- $\exists m \in X, \forall x \in X, m \leq x$
- $\exists M \in X, \forall x \in X, x \leq M$

We will show that $P(n)$ is true for all $n \in \mathbb{N}$.

- **Base case:** Let $X = \{a_1\}$ be given.
  - $\forall x \in X, a_1 \leq x$
  - $\forall x \in X, x \leq a_1$

  Therefore, $P(1)$ is true.

- **Inductive step:** Let $P(n)$ be true for $n \in \mathbb{N}$. Let
  - $X = \{a_1, a_2, \ldots, a_{n+1}\}$
  - $Y = \{a_1, a_2, \ldots, a_n\} \subseteq X$

  be given. Then
  - $\exists k \in \mathbb{N}^n, \forall y \in Y, a_k \leq y$
  - $\exists K \in \mathbb{N}^n, \forall y \in Y, y \leq a_K$

  Also,
  - $\forall x \in X, \min\{a_k, a_{n+1}\} \leq x$
  - $\forall x \in X, x \leq \max\{a_K, a_{n+1}\}$

Therefore, $P(n+1)$ is true.

By the *principle of mathematical induction*, $P(n)$ is true for all $n \in \mathbb{N}$. $\square$

Lemma 2.2.6. *Let $n, m \in \mathbb{N}_0$ be given. Then*

(A) $(\exists f : \mathbb{N}^n \hookrightarrow \mathbb{N}^m) \implies n \leq m$

(B) $(\exists f : \mathbb{N}^n \twoheadrightarrow \mathbb{N}^m) \implies n \geq m$

(C) $\left(\exists f : \mathbb{N}^n \xrightarrow{\sim} \mathbb{N}^m\right) \implies n = m$

(C) *guarantees the uniqueness of the cardinality.*

$$\begin{gathered} n = |X| \wedge m = |X| \\ \Downarrow \\ \mathbb{N}^n \sim X \wedge \mathbb{N}^m \sim X \\ \Downarrow \\ \mathbb{N}^n \sim \mathbb{N}^m \\ \Downarrow \\ n = m \end{gathered}$$

Proof. **(A)** Define

$$P(n) : \forall m \in \mathbb{N}_0, (\exists f_n : \mathbb{N}^n \hookrightarrow \mathbb{N}^m) \implies n \leq m$$

We will show that $P(n)$ is true for all $n \in \mathbb{N}_0$.

- **Base case:**
$$\begin{gathered} \forall m \in \mathbb{N}_0, 0 \leq m \\ \Downarrow \\ P(0) \text{ is true} \end{gathered}$$

- **Inductive step:** Let $P(n)$ be true for $n \in \mathbb{N}_0$. If $m = 0$, then
$$\nexists f_{n+1} : \mathbb{N}^{n+1} \hookrightarrow \mathbb{N}^0$$
Otherwise, let $f_{n+1} : \mathbb{N}^{n+1} \hookrightarrow \mathbb{N}^m$ be given with
$$f_{n+1}(n+1) = x \in \mathbb{N}^m$$
Define $g_m : \mathbb{N}^m \xrightarrow{\sim} \mathbb{N}^m$ as
$$\forall y \in \mathbb{N}^m, g_m(y) = \begin{cases} x, & \text{if } y = m \\ m, & \text{if } y = x \\ y, & \text{otherwise} \end{cases}$$

Define $f_n : \mathbb{N}^n \to \mathbb{N}^{m-1}$ as

$$f_n \equiv (g_m \circ f_{n+1})|_{\mathbb{N}^n}^{\mathbb{N}^{m-1}}$$

By Theorems 2.1.7 and 2.1.9,

$$\begin{gathered} f_n \text{ is injective} \\ \Downarrow \\ n \leq m-1 \\ \Downarrow \\ n+1 \leq m \\ \Downarrow \\ P(n+1) \text{ is true} \end{gathered}$$

By the *principle of mathematical induction*, $P(n)$ is true for all $n \in \mathbb{N}_0$.

(B) Define

$$Q(m) : \forall n \in \mathbb{N}_0, (\exists f_m : \mathbb{N}^n \twoheadrightarrow \mathbb{N}^m) \implies n \geq m$$

We will show that $Q(m)$ is true for all $m \in \mathbb{N}_0$.

- **Base case:**

$$\begin{gathered} \forall n \in \mathbb{N}_0, n \geq 0 \\ \Downarrow \\ Q(0) \text{ is true} \end{gathered}$$

- **Inductive step:** Let $Q(m)$ be true for $m \in \mathbb{N}_0$. If $n = 0$, then

$$\nexists f_{m+1} : \mathbb{N}^0 \twoheadrightarrow \mathbb{N}^{m+1}$$

Otherwise, let $f_{m+1} : \mathbb{N}^n \twoheadrightarrow \mathbb{N}^{m+1}$ be given with

$$f_{m+1}(n) = x \in \mathbb{N}^{m+1}$$

Define $g_{m+1} : \mathbb{N}^{m+1} \xrightarrow{\sim} \mathbb{N}^{m+1}$ as

$$\forall y \in \mathbb{N}^{m+1}, g_{m+1}(y) = \begin{cases} x, & \text{if } y = m+1 \\ m+1, & \text{if } y = x \\ y, & \text{otherwise} \end{cases}$$

Define

$$K_{m+1} \coloneqq (g_{m+1} \circ f_{m+1})^{-1}(\{m+1\}) \subseteq \mathbb{N}^n$$

Define $f_m : \mathbb{N}^{n-1} \to \mathbb{N}^m$ as

$$\forall y \in \mathbb{N}^{n-1}, f_m(y) = \begin{cases} (g_{m+1} \circ f_{m+1})(y), & \text{if } y \in \mathbb{N}^{n-1} \setminus K_{m+1} \\ m, & \text{if } y \in K_{m+1} \end{cases}$$

By Theorem 2.1.9,

$$\begin{gathered} f_m \text{ is surjective} \\ \Downarrow \\ n-1 \geq m \\ \Downarrow \\ n \geq m+1 \\ \Downarrow \\ Q(m+1) \text{ is true} \end{gathered}$$

By the *principle of mathematical induction*, $Q(m)$ is true for all $m \in \mathbb{N}_0$.

(C) follows from **(A)** and **(B)**.

□

Lemma 2.2.7. *Let $n, m \in \mathbb{N}_0$ be given. Then*

(A) $n \leq m \implies \exists f : \mathbb{N}^n \hookrightarrow \mathbb{N}^m$

(B) $n \geq m \implies \exists f : \mathbb{N}^n \twoheadrightarrow \mathbb{N}^m$

(C) $n = m \implies \exists f : \mathbb{N}^n \xrightarrow{\sim} \mathbb{N}^m$

This lemma is the converse of Lemma 2.2.6.

Proof. **(A)** If $n \leq m$, define $f : \mathbb{N}^n \hookrightarrow \mathbb{N}^m$ as

$$\forall i \in \mathbb{N}^n, f(i) = i$$

(B) If $n \geq m$, define $f : \mathbb{N}^n \to \mathbb{N}^m$ as

$$\forall i \in \mathbb{N}^n, f(i) = \begin{cases} i, & \text{if } i \leq m \\ m, & \text{if } i > m \end{cases}$$

(C) follows from **(A)** and **(B)**.

□

Theorem 2.2.8. *Let X, Y be finite. Then*

(A) $|X| \leq |Y| \iff \exists f : X \hookrightarrow Y$

(B) $|X| \geq |Y| \iff \exists f : X \twoheadrightarrow Y$

(C) $|X| = |Y| \iff \exists f : X \xrightarrow{\sim} Y$

The contraposition of ***(A,*** $\Longleftarrow$ *) is called the* ***pigeonhole principle****.*

Proof. Since X and Y are finite,

- $\exists n \in \mathbb{N}_0, \exists f_X : \mathbb{N}^n \xrightarrow{\sim} X$
- $\exists m \in \mathbb{N}_0, \exists f_Y : \mathbb{N}^m \xrightarrow{\sim} Y$

($\Longleftarrow$) Define $f : X \to Y$ and $F : \mathbb{N}^n \to \mathbb{N}^m$ as

$$F \equiv (f_Y)^{-1} \circ f \circ f_X$$

By Theorems 2.1.5 and 2.1.9,

- f is injective $\implies$ F is injective
- f is surjective $\implies$ F is surjective

By Lemma 2.2.6,

(A) $(\exists f : X \hookrightarrow Y) \implies (\exists F : \mathbb{N}^n \hookrightarrow \mathbb{N}^m) \implies n \leq m$

(B) $(\exists f : X \twoheadrightarrow Y) \implies (\exists F : \mathbb{N}^n \twoheadrightarrow \mathbb{N}^m) \implies n \geq m$

(C) follows from **(A)** and **(B)**.

($\implies$) Define $f : X \to Y$ and $G : \mathbb{N}^m \to \mathbb{N}^n$ as

$$f \equiv f_Y \circ G \circ (f_X)^{-1}$$

By Theorems 2.1.5 and 2.1.9,

(A) G is injective $\implies$ f is injective

(B) G is surjective $\implies$ f is surjective

By Lemma 2.2.7,

(A) $n \leq m \implies (\exists G : \mathbb{N}^n \hookrightarrow \mathbb{N}^m) \implies (\exists f : X \hookrightarrow Y)$

(B) $n \geq m \implies (\exists G : \mathbb{N}^n \twoheadrightarrow \mathbb{N}^m) \implies (\exists f : X \twoheadrightarrow Y)$

(C) follows from **(A)** and **(B)**.

□

Lemma 2.2.9.

$$\forall X \forall Y, |X| < \aleph_0 \wedge |Y| < \aleph_0 \wedge X \cap Y = \varnothing \implies |X \cup Y| < \aleph_0$$

Proof. Assume that

$$|X| < \aleph_0 \wedge |Y| < \aleph_0 \wedge X \cap Y = \varnothing$$

Then

- $\exists n \in \mathbb{N}_0, \exists f : \mathbb{N}^n \xrightarrow{\sim} X$
- $\exists m \in \mathbb{N}_0, \exists g : \mathbb{N}^m \xrightarrow{\sim} Y$

Define $h : \mathbb{N}^{m+n} \xrightarrow{\sim} X \cup Y$ as

$$\forall i \in \mathbb{N}^{m+n}, h(i) = \begin{cases} f(i), & \text{if } i \leq n \\ g(i-n), & \text{if } n < i \leq n+m \end{cases}$$

Then

$$|X \cup Y| = m + n < \aleph_0$$

□

Theorem 2.2.10.

$$\forall X \forall Y, |X| < \aleph_0 \wedge |Y| < \aleph_0 \implies |X \cup Y| < \aleph_0$$

Proof. Assume that

$$|X| < \aleph_0 \wedge |Y| < \aleph_0$$

Define

- $A = X \setminus Y \subseteq X$
- $B = Y \setminus X \subseteq Y$
- $C = X \cap Y \subseteq X$

By Theorem 2.2.2,

- $|A| < \aleph_0$, $|B| < \aleph_0$, and $|C| < \aleph_0$
- $A \cap B = \varnothing$, $B \cap C = \varnothing$, and $C \cap A = \varnothing$

Consider the following:

- $X = (X \setminus Y) \cup (X \cap Y) = A \cup C$
- $Y = (Y \setminus X) \cup (X \cap Y) = B \cup C$
- $X \cup Y = ((X \setminus Y) \cup (Y \setminus X)) \cup (X \cap Y) = (A \cup B) \cup C$

By Lemma 2.2.9,

$$\begin{gathered} |A \cup B| < \aleph_0 \\ \Downarrow \\ |(A \cup B) \cup C| < \aleph_0 \\ \Downarrow \\ |X \cup Y| < \aleph_0 \end{gathered}$$

□

Theorem 2.2.11.

$$\forall n \in \mathbb{N}, (\forall k \in \mathbb{N}^n, |X_k| < \aleph_0) \implies \left|\bigcup_{k=1}^{n} X_k\right| < \aleph_0$$

Proof. Define

$$P(n) : (\forall k \in \mathbb{N}^n, |X_k| < \aleph_0) \implies \left|\bigcup_{k=1}^{n} X_k\right| < \aleph_0$$

We will show that $P(n)$ is true for all $n \in \mathbb{N}$.

- **Base case:** Let $|X_1| < \aleph_0$ be given. Then

$$\begin{gathered} |X_1| < \aleph_0 \\ \Downarrow \\ \left|\bigcup_{k=1}^{1} X_k\right| < \aleph_0 \\ \Downarrow \\ P(1) \text{ is true} \end{gathered}$$

- **Inductive step:** Let $P(n)$ be true for $n \in \mathbb{N}$. Assume that

$$\forall k \in \mathbb{N}^{n+1}, |X_k| < \aleph_0$$

  By the inductive hypothesis,

  – $\left|\bigcup_{k=1}^{n} X_k\right| < \aleph_0$
  – $|X_{n+1}| < \aleph_0$

  By Theorem 2.2.10,

$$\begin{gathered} \bigcup_{k=1}^{n+1} X_k = \left(\bigcup_{k=1}^{n} X_k\right) \cup X_{n+1} \\ \Downarrow \\ \left|\bigcup_{k=1}^{n+1} X_k\right| < \aleph_0 \\ \Downarrow \\ P(n+1) \text{ is true} \end{gathered}$$

By the *principle of mathematical induction*, $P(n)$ is true for all $n \in \mathbb{N}$. □

Lemma 2.2.12.

$$\forall X \forall Y, |X| < \aleph_0 \wedge |Y| < \aleph_0 \implies |X \times Y| < \aleph_0$$

Proof. Assume that

$$|X| < \aleph_0 \wedge |Y| < \aleph_0$$

Then

- $\exists n \in \mathbb{N}_0, \exists f : \mathbb{N}^n \xrightarrow{\sim} X$
- $\exists m \in \mathbb{N}_0, \exists g : \mathbb{N}^m \xrightarrow{\sim} Y$

Define

$$i = (a-1)m + b \in \mathbb{N}^{nm}$$

for each $a \in \mathbb{N}^n$ and $b \in \mathbb{N}^m$. Define $h : \mathbb{N}^{nm} \xrightarrow{\sim} X \times Y$ as

$$\forall i \in \mathbb{N}^{nm}, h(i) = (f(a), g(b))$$

Then

$$|X \times Y| = nm < \aleph_0$$

□

Theorem 2.2.13.

$$\forall n \in \mathbb{N}, (\forall k \in \mathbb{N}^n, |X_k| < \aleph_0) \implies \left| \prod_{k=1}^{n} X_k \right| < \aleph_0$$

Proof. Define

$$P(n) : (\forall k \in \mathbb{N}^n, |X_k| < \aleph_0) \implies \left| \prod_{k=1}^{n} X_k \right| < \aleph_0$$

We will show that $P(n)$ is true for all $n \in \mathbb{N}$.

- **Base case:** Let $|X_1| < \aleph_0$ be given. Then

$$\begin{gathered} |X_1| < \aleph_0 \\ \Downarrow \\ \left| \prod_{k=1}^{1} X_k \right| < \aleph_0 \\ \Downarrow \\ P(1) \text{ is true} \end{gathered}$$

- **Inductive step:** Let $P(n)$ be true for $n \in \mathbb{N}$. Assume that

$$\forall k \in \mathbb{N}^{n+1}, |X_k| < \aleph_0$$

By the inductive hypothesis,

  - $\left|\prod_{k=1}^{n} X_k\right| < \aleph_0$
  - $|X_{n+1}| < \aleph_0$

By Lemma 2.2.12,

$$\begin{gathered}\prod_{k=1}^{n+1} X_k \cong \left(\prod_{k=1}^{n} X_k\right) \times X_{n+1} \\ \Downarrow \\ \left|\prod_{k=1}^{n+1} X_k\right| < \aleph_0 \\ \Downarrow \\ P(n+1) \text{ is true}\end{gathered}$$

By the *principle of mathematical induction*, $P(n)$ is true for all $n \in \mathbb{N}$. □

Theorem 2.2.14.

$$\forall X, \nexists f : X \twoheadrightarrow \mathcal{P}(X)$$

where

$$\mathcal{P}(X) \coloneqq \{E \mid E \subseteq X\}$$

is called the ***power set*** *of X. This theorem is called* ***Cantor's theorem****.*

Proof. Assume that

$$\exists f : X \twoheadrightarrow \mathcal{P}(X)$$

Define

$$S \coloneqq \{x \in X \mid x \notin f(x)\} \subseteq X$$

Then

$$\begin{gathered}S \subseteq X \\ \Downarrow \\ S \in \mathcal{P}(X) \\ \Downarrow \\ \exists x \in X, f(x) = S\end{gathered}$$

There are two cases:

- If $x \in S$, then

$$\begin{gathered} x \in S \\ \Downarrow \\ x \notin f(x) = S \\ \Downarrow \\ \perp \end{gathered}$$

- If $x \notin S$, then

$$\begin{gathered} x \notin S \\ \Downarrow \\ x \in f(x) = S \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$\nexists f : X \twoheadrightarrow \mathcal{P}(X)$$

□

2.3 Countable and Uncountable Sets

Theorem 2.3.1.

$$\forall X \forall Y, |X| \geq \aleph_0 \wedge |Y| < \aleph_0 \implies |X \setminus Y| \geq \aleph_0$$

Proof. Assume that

$$\exists X \exists Y, |X| \geq \aleph_0 \wedge |Y| < \aleph_0 \wedge |X \setminus Y| < \aleph_0$$

By Theorem 2.2.10,

$$|(X \setminus Y) \cup Y| < \aleph_0$$

By Theorem 2.2.2,

$$\begin{gathered} X = (X \setminus Y) \cup (X \cap Y) \\ \Downarrow \\ X \subseteq (X \setminus Y) \cup Y \\ \Downarrow \\ |X| < \aleph_0 \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$\forall X \forall Y, |X| \geq \aleph_0 \wedge |Y| < \aleph_0 \implies |X \setminus Y| \geq \aleph_0$$

□

Theorem 2.3.2.

$$\forall X \forall Y, |X| \geq \aleph_0 \wedge (\exists f : X \hookrightarrow Y) \implies |Y| \geq \aleph_0$$

Proof. Assume that

$$\exists X \exists Y, |X| \geq \aleph_0 \wedge (\exists f : X \hookrightarrow Y) \wedge |Y| < \aleph_0$$

Define $g : Y \twoheadrightarrow X$ as

$$\forall y \in Y, g(y) = \begin{cases} f^{-1}(y), & \text{if } y \in f(X) \\ f^{-1}(y_0), & \text{otherwise} \end{cases}$$

where $y_0 \in f(X)$ is fixed. By Theorem 2.2.4,

$$\begin{gathered} \exists g : Y \twoheadrightarrow X \\ \Downarrow \\ |X| < \aleph_0 \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$\forall X \forall Y, |X| \geq \aleph_0 \wedge (\exists f : X \hookrightarrow Y) \implies |Y| \geq \aleph_0$$

□

Lemma 2.3.3.

$$|\mathbb{N}| \geq \aleph_0$$

Proof. Assume that

$$|\mathbb{N}| < \aleph_0$$

By Theorem 2.2.5,

$$\exists m \in \mathbb{N}, \forall x \in \mathbb{N}, x \leq m$$

Then

$$\begin{gathered} m < m + 1 \\ \Downarrow \\ m + 1 \notin \mathbb{N} \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$|\mathbb{N}| \geq \aleph_0$$

□

Theorem 2.3.4.

$$\forall X, |X| = \aleph_0 \implies |X| \geq \aleph_0$$

Proof. Let $|X| = \aleph_0$ be given. Then

$$\exists f : \mathbb{N} \xrightarrow{\sim} X$$

By Theorem 2.3.2 and lemma 2.3.3,

$$|X| \geq \aleph_0$$

$\square$

Theorem 2.3.5.

$$|\mathbb{Z}| = \aleph_0$$

Proof. Define $f : \mathbb{N} \to \mathbb{Z}$ as

$$\forall n \in \mathbb{N}, f(n) = \begin{cases} -\dfrac{n}{2}, & \text{if } n \text{ is even} \\ \dfrac{n-1}{2}, & \text{if } n \text{ is odd} \end{cases}$$

We will show that

$$\begin{gathered} f \text{ is bijective} \\ \Downarrow \\ |\mathbb{Z}| = \aleph_0 \end{gathered}$$

- **Injectivity:** Let $f(n_1) = f(n_2)$ be given for $n_1, n_2 \in \mathbb{N}$. Since
  - n is even $\implies f(n) = -\dfrac{n}{2} < 0$
  - n is odd $\implies f(n) = \dfrac{n-1}{2} \geq 0$

  n_1 and n_2 are both even or both odd.
  - If both n_1 and n_2 are even, then

    $$\begin{gathered} -\frac{n_1}{2} = -\frac{n_2}{2} \\ \Downarrow \\ n_1 = n_2 \end{gathered}$$

  - If both n_1 and n_2 are odd, then

    $$\begin{gathered} \frac{n_1 - 1}{2} = \frac{n_2 - 1}{2} \\ \Downarrow \\ n_1 = n_2 \end{gathered}$$

- **Surjectivity:** Let $k \in \mathbb{Z}$ be given.
  - If $k \geq 0$, then

$$\begin{gathered} 2k+1 \in \mathbb{N} \\ \Downarrow \\ f(2k+1) = \frac{(2k+1)-1}{2} = \frac{2k}{2} = k \end{gathered}$$

  - If $k < 0$, then

$$\begin{gathered} -2k \in \mathbb{N} \\ \Downarrow \\ f(-2k) = -\frac{-2k}{2} = k \end{gathered}$$

$\square$

Lemma 2.3.6.

$$\forall X \subseteq \mathbb{N}, |X| \geq \aleph_0 \implies |X| = \aleph_0$$

Proof. Assume that

- $X \subseteq \mathbb{N}$
- $|X| \geq \aleph_0$

Define $(X_n)_{n=1}^{\infty}$ and $f : \mathbb{N} \to X$ as

$$\begin{array}{cc} X_1 = X \subseteq \mathbb{N} & f(1) = \min X_1 \\ X_2 = X_1 \setminus \{f(1)\} \subseteq \mathbb{N} & f(2) = \min X_2 \\ \vdots & \vdots \\ X_{n+1} = X_n \setminus \{f(n)\} \subseteq \mathbb{N} & f(n+1) = \min X_{n+1} \end{array}$$

for each $n \in \mathbb{N}$. By Theorems 1.1.4 and 2.3.1,

$$\begin{gathered} \forall n \in \mathbb{N}, |X_n| \geq \aleph_0 \\ \Downarrow \\ \forall n \in \mathbb{N}, X_n \neq \varnothing \\ \Downarrow \\ \forall n \in \mathbb{N}, \exists f(n) = \min X_n \in X_n \\ \Downarrow \\ f \text{ is well-defined} \end{gathered}$$

We will show that

$$\begin{gathered} f \text{ is bijective} \\ \Downarrow \\ |X| = \aleph_0 \end{gathered}$$

- **Injectivity:** Let $n_1 \neq n_2$ (*WLOG* $n_1 < n_2$) be given for $n_1, n_2 \in \mathbb{N}$.
  - $f(n_1) = \min X_{n_1} \in X_{n_1}$
  - $f(n_2) = \min X_{n_2} \in X_{n_2}$

  Then
  $$\begin{gathered} X_{n_2} \subseteq X_{n_1} \setminus \{f(n_1)\} \\ \Downarrow \\ f(n_2) \neq f(n_1) \\ \Downarrow \\ f \text{ is injective} \end{gathered}$$

- **Surjectivity:** Let $x \in X$ be given. Define
  $$S \coloneqq \{n \in X \mid n < x\}$$
  By Lemma 2.2.1,
  $$\begin{gathered} S \subseteq \mathbb{N}^{x-1} \\ \Downarrow \\ |S| < \aleph_0 \\ \Downarrow \\ \exists k \in \mathbb{N}_0, |S| = k \end{gathered}$$
  If $k = 0$, then
  - $S = \varnothing$
  - $x = \min X_1 = f(1)$

  Otherwise,
  - $S = \{s_1, s_2, \ldots, s_k\}$
  - $s_1 < s_2 < \cdots < s_k$

  Then
  $$\begin{gathered} \min X_1 = \min S \\ \min X_2 = \min S \setminus \{s_1\} \\ \vdots \\ \min X_k = \min S \setminus \{s_1, s_2, \ldots, s_{k-1}\} \end{gathered} \qquad \begin{gathered} X_2 = X \setminus \{s_1\} \\ X_3 = X \setminus \{s_1, s_2\} \\ \vdots \\ X_{k+1} = X \setminus S \end{gathered}$$
  Therefore,
  $$\begin{gathered} f(k+1) = \min X_{k+1} = x \\ \Downarrow \\ f \text{ is surjective} \end{gathered}$$

$\square$

Theorem 2.3.7.

$$\forall X, \forall Y \subseteq X, |X| = \aleph_0 \wedge |Y| \geq \aleph_0 \implies |Y| = \aleph_0$$

Proof. Assume that

- $Y \subseteq X$
- $|X| = \aleph_0$
- $|Y| \geq \aleph_0$

By Theorem 2.1.7,

$$\begin{gathered} \exists f : \mathbb{N} \xrightarrow{\sim} X \\ \Downarrow \\ \exists f|_{f^{-1}(Y)}^{Y} : f^{-1}(Y) \xrightarrow{\sim} Y \end{gathered}$$

By Theorem 2.2.4, if $f^{-1}(Y)$ is finite, then

$$\begin{gathered} \left|f^{-1}(Y)\right| < \aleph_0 \\ \Downarrow \\ |Y| < \aleph_0 \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$\left|f^{-1}(Y)\right| \geq \aleph_0$$

By Lemma 2.3.6,

$$\begin{gathered} \left|f^{-1}(Y)\right| \subseteq \mathbb{N} \\ \Downarrow \\ \left|f^{-1}(Y)\right| = \aleph_0 \\ \Downarrow \\ \exists g : \mathbb{N} \xrightarrow{\sim} f^{-1}(Y) \\ \Downarrow \\ \exists f|_{f^{-1}(Y)}^{Y} \circ g : \mathbb{N} \xrightarrow{\sim} Y \\ \Downarrow \\ |Y| = \aleph_0 \end{gathered}$$

□

Theorem 2.3.8.

$$\forall X, |X| \geq \aleph_0 \wedge (\exists f : \mathbb{N} \twoheadrightarrow X) \implies |X| = \aleph_0$$

Proof. Assume that

- $|X| \geq \aleph_0$
- $\exists f : \mathbb{N} \twoheadrightarrow X$

Define
$$S_k \coloneqq f^{-1}(\{k\})$$
for each $k \in X$. By Theorem 1.1.4,
$$\begin{gathered} f \text{ is surjective} \\ \Downarrow \\ \forall k \in X, S_k \neq \varnothing \\ \Downarrow \\ \forall k \in X, \exists m \in S_k, m = \min S_k \end{gathered}$$
Define $g : X \to \mathbb{N}$ as
$$\forall k \in X, g(k) = \min S_k$$
Let $k_1, k_2 \in X$ be given. Then
$$\begin{gathered} \forall i, j \in X, i \neq j \implies S_i \cap S_j = \varnothing \\ \Downarrow \\ g(k_1) = g(k_2) \implies \min S_{k_1} = \min S_{k_2} \implies k_1 = k_2 \\ \Downarrow \\ g \text{ is injective} \end{gathered}$$
By Theorem 2.1.7,
$$\exists g|^{g(X)} : X \xrightarrow{\sim} g(X)$$
By Theorem 2.3.2,
$$\begin{gathered} |X| \geq \aleph_0 \\ \Downarrow \\ |g(X)| \geq \aleph_0 \end{gathered}$$
By Lemma 2.3.6,
$$\begin{gathered} g(X) \subseteq \mathbb{N} \\ \Downarrow \\ |g(X)| = \aleph_0 \\ \Downarrow \\ |X| = \aleph_0 \end{gathered}$$
□

Theorem 2.3.9.
$$(\forall n \in \mathbb{N}, |X_n| < \aleph_0) \implies \left|\bigcup_{n=1}^{\infty} X_n\right| \leq \aleph_0$$

Proof. Assume that

$$\begin{gathered}\forall n \in \mathbb{N}, |X_n| < \aleph_0 \\ \Downarrow \\ \forall n \in \mathbb{N}, \exists m_n \in \mathbb{N}_0, \exists g_n : \mathbb{N}^{m_n} \xrightarrow{\sim} X_n\end{gathered}$$

If

$$\left|\bigcup_{n=1}^{\infty} X_n\right| < \aleph_0$$

then the proof is complete. Otherwise,

$$\left|\bigcup_{n=1}^{\infty} X_n\right| \geq \aleph_0$$

Define $h : \mathbb{N} \twoheadrightarrow \displaystyle\bigcup_{n=1}^{\infty} X_n$ as

- $\forall m \in \mathbb{N}_{m_1}, h(m) = g_1(m)$
- $\forall n \in \mathbb{N}_2, \forall m \in \mathbb{N}_{m_n}, h\left(\displaystyle\sum_{k=1}^{n-1} m_i + m\right) = g_n(m)$

By Theorem 2.3.8,

$$\begin{gathered}\left|\bigcup_{n=1}^{\infty} X_n\right| \geq \aleph_0 \\ \Downarrow \\ \left|\bigcup_{n=1}^{\infty} X_n\right| = \aleph_0\end{gathered}$$

□

Theorem 2.3.10.

$$(\forall n \in \mathbb{N}, |X_n| = \aleph_0) \implies \left|\bigcup_{n=1}^{\infty} X_n\right| = \aleph_0$$

Proof. Assume that

$$\left|\bigcup_{n=1}^{\infty} X_n\right| < \aleph_0$$

By Theorem 2.2.2,

$$\begin{gathered} X_1 \subseteq \bigcup_{n=1}^{\infty} X_n \\ \Downarrow \\ |X_1| < \aleph_0 \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$\left| \bigcup_{n=1}^{\infty} X_n \right| \geq \aleph_0$$

Define

$$T_n := \left\{ k \in \mathbb{N} \,\middle|\, n \leq \frac{k(k+1)}{2} \right\}$$

for each $n \in \mathbb{N}$. By Theorem 1.1.4,

$$\begin{gathered} 2n \in T_n \\ \Downarrow \\ T_n \neq \varnothing \\ \Downarrow \\ \exists t_n \in T_n, t_n = \min T_n \end{gathered}$$

for all $n \in \mathbb{N}$. Define

$$s_n := n - \frac{t_n(t_n - 1)}{2} = n - \frac{t_n(t_n + 1)}{2} + t_n \in \mathbb{N}$$

for each $n \in \mathbb{N}$. Then

$$\begin{gathered} \frac{t_n(t_n - 1)}{2} < n \leq \frac{t_n(t_n + 1)}{2} \\ \Downarrow \\ t_n - s_n + 1 = \frac{t_n(t_n + 1)}{2} - n + 1 \geq 1 \\ \Downarrow \\ t_n - s_n + 1 \in \mathbb{N} \end{gathered}$$

for all $n \in \mathbb{N}$. Assume that

$$\begin{gathered} \forall n \in \mathbb{N}, |X_n| = \aleph_0 \\ \Downarrow \\ \forall n \in \mathbb{N}, \exists f_n : \mathbb{N} \xrightarrow{\sim} X_n \end{gathered}$$

Define $g : \mathbb{N} \to \bigcup_{n=1}^{\infty} X_n$ as

$$\forall n \in \mathbb{N}, g(n) = f_{s_n}(t_n - s_n + 1)$$

Let $x \in \bigcup_{n=1}^{\infty} X_n$ be given. Then

$$\begin{gathered}\exists m \in \mathbb{N}, x \in X_m \\ \Downarrow \\ \exists l \in \mathbb{N}, f_m(l) = x\end{gathered}$$

Define

$$k := \frac{(l+m-1)(l+m-2)}{2} + m \in \mathbb{N}$$

We will show that

$$\begin{gathered}l+m-2 \notin T_k \wedge l+m-1 \in T_k \\ \Downarrow \\ t_k = \min T_k = l+m-1 \\ \Downarrow \\ s_k = k - \frac{t_k(t_k-1)}{2} = m \\ \Downarrow \\ g(k) = f_{s_k}(t_k - s_k + 1) = f_m(l+m-1-m+1) = f_m(l) = x\end{gathered}$$

Consider the following inequalities:

- Let $z = l+m-2$ be given. Then

$$\begin{aligned}k - \frac{z(z+1)}{2} &= k - \frac{(l+m-2)(l+m-1)}{2} \\ &= \frac{(l+m-1)(l+m-2)}{2} + m - \frac{(l+m-2)(l+m-1)}{2} \\ &= m > 0\end{aligned}$$

  Therefore,

$$z = l+m-2 \notin T_k$$

- Let $z = l+m-1$ be given. Then

$$\begin{aligned}k - \frac{z(z+1)}{2} &= k - \frac{(l+m-1)(l+m)}{2} \\ &= \frac{(l+m-1)(l+m-2)}{2} + m - \frac{(l+m-1)(l+m)}{2} \\ &= m - \frac{(l+m-1)(l+m) - (l+m-1)(l+m-2)}{2} \\ &= m - (l+m-1) = 1 - l \le 0\end{aligned}$$

  Therefore,

$$z = l+m-1 \in T_k$$

By Theorem 2.3.8,

$$\begin{gathered} g \text{ is surjective} \\ \Downarrow \\ \left| \bigcup_{n=1}^{\infty} X_n \right| = \aleph_0 \end{gathered}$$

□

Corollary 2.3.11.

$$\forall n \in \mathbb{N}, (\forall k \in \mathbb{N}^n, |X_k| = \aleph_0) \implies \left| \bigcup_{k=1}^{n} X_k \right| = \aleph_0$$

Proof. Assume that

$$(\forall k \in \mathbb{N}^n, |X_k| = \aleph_0)$$

for $n \in \mathbb{N}$. Define

$$X_k \coloneqq X_n$$

for all $k \in \mathbb{N}_{n+1}$. By Theorem 2.3.10,

$$\begin{gathered} \forall n \in \mathbb{N}, |X_n| = \aleph_0 \\ \Downarrow \\ \left| \bigcup_{n=1}^{\infty} X_n \right| = \aleph_0 \end{gathered}$$

Therefore,

$$\begin{gathered} \bigcup_{k=1}^{n} X_k = \bigcup_{n=1}^{\infty} X_n \\ \Downarrow \\ \left| \bigcup_{k=1}^{n} X_k \right| = \aleph_0 \end{gathered}$$

□

Lemma 2.3.12.

$$|\mathbb{N} \times \mathbb{N}| = \aleph_0$$

Proof. Define $f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ as

$$\forall m, n \in \mathbb{N}, f(m, n) = \frac{(m+n-1)(m+n-2)}{2} + m$$

We will show that f is bijective.

- **Injectivity:** Let $(m_1, n_1) \neq (m_2, n_2)$ be given for $m_1, n_1, m_2, n_2 \in \mathbb{N}$. If

$$m_1 + n_1 = m_2 + n_2$$

then

$$\begin{gathered} m_1 \neq m_2 \\ \Downarrow \\ f(m_1, n_1) \neq f(m_2, n_2) \end{gathered}$$

Otherwise, $m_1 + n_1 \neq m_2 + n_2$ (*WLOG* $m_1 + n_1 < m_2 + n_2$). Then

$$\begin{aligned} f(m_1, n_1) &= \frac{(m_1 + n_1 - 1)(m_1 + n_1 - 2)}{2} + m_1 \\ &\leq \frac{(m_1 + n_1 - 1)(m_1 + n_1 - 2)}{2} + (m_1 + n_1 - 1) \\ &= \frac{(m_1 + n_1)(m_1 + n_1 - 1)}{2} \\ &\leq \frac{(m_2 + n_2 - 1)(m_2 + n_2 - 2)}{2} \\ &< \frac{(m_2 + n_2 - 1)(m_2 + n_2 - 2)}{2} + m_2 = f(m_2, n_2) \end{aligned}$$

- **Surjectivity:** Let $k \in \mathbb{N}$ be given. Define t_k, s_k as in the proof of Theorem 2.3.10. Then

$$f(s_k, t_k - s_k + 1) = k$$

Therefore,

$$|\mathbb{N} \times \mathbb{N}| = \aleph_0$$

□

Theorem 2.3.13.

$$\forall X \forall Y, |X| = \aleph_0 \wedge |Y| = \aleph_0 \implies |X \times Y| = \aleph_0$$

Proof. Since X and Y are countable,

- $\exists f : \mathbb{N} \xrightarrow{\sim} X$
- $\exists g : \mathbb{N} \xrightarrow{\sim} Y$

Define $h : \mathbb{N} \times \mathbb{N} \xrightarrow{\sim} X \times Y$ as

$$\forall m, n \in \mathbb{N}, h(m, n) = (f(m), g(n))$$

By Lemma 2.3.12,

$$|X \times Y| = \aleph_0$$

□

Theorem 2.3.14.

$$\forall n \in \mathbb{N}, (\forall k \in \mathbb{N}^n, |X_k| = \aleph_0) \implies \left|\prod_{k=1}^{n} X_k\right| = \aleph_0$$

Proof. Define

$$P(n) : (\forall k \in \mathbb{N}^n, |X_k| = \aleph_0) \implies \left|\prod_{k=1}^{n} X_k\right| = \aleph_0$$

We will show that $P(n)$ is true for all $n \in \mathbb{N}$.

- **Base case:** Let $|X_1| = \aleph_0$ be given. Then

$$\begin{gathered} |X_1| = \aleph_0 \\ \Downarrow \\ \left|\prod_{k=1}^{1} X_k\right| = \aleph_0 \\ \Downarrow \\ P(1) \text{ is true} \end{gathered}$$

- **Inductive step:** Let $P(n)$ be true for $n \in \mathbb{N}$. Assume that

$$\forall k \in \mathbb{N}^{n+1}, |X_k| = \aleph_0$$

  By the inductive hypothesis,

  - $\left|\prod_{k=1}^{n} X_k\right| = \aleph_0$
  - $|X_{n+1}| = \aleph_0$

  By Theorem 2.3.13,

$$\begin{gathered} \prod_{k=1}^{n+1} X_k \cong \left(\prod_{k=1}^{n} X_k\right) \times X_{n+1} \\ \Downarrow \\ \left|\prod_{k=1}^{n+1} X_k\right| = \aleph_0 \\ \Downarrow \\ P(n+1) \text{ is true} \end{gathered}$$

By the *principle of mathematical induction*, $P(n)$ is true for all $n \in \mathbb{N}$. □

Theorem 2.3.15.

$$|\mathbb{Q}| = \aleph_0$$

Proof. Let $q \in \mathbb{Q}$ be given. Then

$$\exists m \in \mathbb{Z}, \exists n \in \mathbb{N}, q = \frac{m}{n}$$

Define $f : \mathbb{Z} \times \mathbb{N} \twoheadrightarrow \mathbb{Q}$ as

$$\forall (m, n) \in \mathbb{Z} \times \mathbb{N}, f(m, n) = \frac{m}{n}$$

By Theorems 2.3.5 and 2.3.13,

$$\begin{gathered} |\mathbb{Z} \times \mathbb{N}| = \aleph_0 \\ \Downarrow \\ \exists g : \mathbb{N} \xrightarrow{\sim} \mathbb{Z} \times \mathbb{N} \end{gathered}$$

By Lemma 2.3.3 and theorem 2.2.2,

$$\begin{gathered} \mathbb{N} \subseteq \mathbb{Q} \\ \Downarrow \\ |\mathbb{Q}| \geq \aleph_0 \end{gathered}$$

By Theorems 2.1.9 and 2.3.8,

$$\begin{gathered} \exists f \circ g : \mathbb{N} \twoheadrightarrow \mathbb{Q} \\ \Downarrow \\ |\mathbb{Q}| = \aleph_0 \end{gathered}$$

□

Lemma 2.3.16.

$$\left| \prod_{n=1}^{\infty} \{0, 1\} \right| > \aleph_0$$

Proof. Define $(e_n)_{n=1}^{\infty}$ in $\displaystyle\prod_{n=1}^{\infty} \{0, 1\}$ as

$$\forall m \in \mathbb{N}, e_{nm} = \begin{cases} 1, & \text{if } m = n \\ 0, & \text{otherwise} \end{cases}$$

for each $n \in \mathbb{N}$. Define $f : \mathbb{N} \hookrightarrow \displaystyle\prod_{n=1}^{\infty} \{0, 1\}$ as

$$\forall n \in \mathbb{N}, f(n) = e_n$$

By Lemma 2.3.3 and theorem 2.3.2,

$$\left|\prod_{n=1}^{\infty}\{0,1\}\right| \geq \aleph_0$$

If $\prod_{n=1}^{\infty}\{0,1\}$ is countable, then

$$\exists g : \mathbb{N} \xrightarrow{\sim} \prod_{n=1}^{\infty}\{0,1\}$$

Let

- $\prod_{n=1}^{\infty}\{0,1\} = \{a_1, a_2, a_3, \dots\}$
- $a_1 = (a_{11}, a_{12}, a_{13}, \dots)$
- $a_2 = (a_{21}, a_{22}, a_{23}, \dots)$

  $\vdots$

be given, where

$$\forall n \in \mathbb{N}, a_n = g(n)$$

Define $b = (b_1, b_2, b_3, \dots) \in \prod_{n=1}^{\infty}\{0,1\}$ as

$$\forall n \in \mathbb{N}, b_n = 1 - a_{nn}$$

Then

$$\begin{gathered}
\forall n \in \mathbb{N}, b \neq a_n \\
\Downarrow \\
b \notin \{a_1, a_2, a_3, \dots\} = \prod_{n=1}^{\infty}\{0,1\} \\
\Downarrow \\
g \text{ is not surjective} \\
\Downarrow \\
\perp
\end{gathered}$$

Therefore,

$$\left|\prod_{n=1}^{\infty}\{0,1\}\right| > \aleph_0$$

$\square$

Theorem 2.3.17.

$$|\mathbb{R}| > \aleph_0$$

Proof. By Lemma 2.3.3 and theorem 2.2.2,

$$\begin{gathered}\mathbb{N} \subseteq \mathbb{R}\\ \Downarrow\\ |\mathbb{R}| \geq \aleph_0\end{gathered}$$

Define $f : \mathbb{N} \hookrightarrow [0,1)$ as

$$\forall n \in \mathbb{N}, f(n) = \frac{1}{n+1}$$

By Lemma 2.3.3 and theorem 2.3.2,

$$|[0,1)| \geq \aleph_0$$

By Theorem 2.3.7,

$$\begin{gathered}[0,1) \subseteq \mathbb{R}\\ \Downarrow\\ |[0,1)| > \aleph_0 \implies |\mathbb{R}| > \aleph_0\end{gathered}$$

Define $g : [0,1) \to \prod_{n=1}^{\infty}\{0,1\}$ as

$$\forall x \in [0,1), g(x) = (a_n^x)_{n=1}^{\infty}$$

where

$$(a_n)_{n=1}^{\infty} \text{ is the } \textit{greedy base } 2 \textit{ expansion} \text{ of } x$$

Define

$$Y := g([0,1)) \subseteq \prod_{n=1}^{\infty}\{0,1\}$$

By Theorems 1.6.9 and 2.1.7,

$$\exists g|^Y : [0,1) \xrightarrow{\sim} Y$$

If $[0,1)$ is countable, then

$$\exists h : \mathbb{N} \xrightarrow{\sim} [0,1)$$

By Theorem 2.1.9,

$$\begin{gathered}g|^Y \circ h : \mathbb{N} \xrightarrow{\sim} Y\\ \Downarrow\\ |Y| = \aleph_0\end{gathered}$$

Define

$$Z := \prod_{n=1}^{\infty}\{0,1\} \setminus Y$$

By Theorems 1.6.10 and 1.6.11,

$$Z = \left\{ (a_n)_{n=1}^{\infty} \in \prod_{n=1}^{\infty} \{0,1\} \,\middle|\, \exists N \in \mathbb{N}, \forall n \in \mathbb{N}_N, a_n = 1 \right\}$$

Define

$$Z_N := \left\{ (a_n)_{n=1}^{\infty} \in \prod_{n=1}^{\infty} \{0,1\} \,\middle|\, \forall n \in \mathbb{N}_N, a_n = 1 \right\} \neq \varnothing$$

for each $N \in \mathbb{N}$. Define $F_N : \prod_{n=1}^{N-1} \{0,1\} \twoheadrightarrow Z_N$ as

$$\forall (a_n)_{n=1}^{N-1} \in \prod_{n=1}^{N-1} \{0,1\}, F_N\left((a_n)_{n=1}^{N-1}\right) = (a_n)_{n=1}^{N-1} + (1)_{n=N}^{\infty}$$

By Theorems 2.2.4 and 2.2.13,

$$\forall N \in \mathbb{N}, |Z_N| < \aleph_0$$

By definition,

$$\begin{gathered} \forall N, M \in \mathbb{N}, N \neq M \implies Z_N \cap Z_M = \varnothing \\ \Downarrow \\ \left| \bigcup_{N=1}^{\infty} Z_N \right| \geq \aleph_0 \end{gathered}$$

By Theorem 2.3.9,

$$\begin{gathered} Z = \bigcup_{N=1}^{\infty} Z_N \\ \Downarrow \\ |Z| = \aleph_0 \end{gathered}$$

By Corollary 2.3.11 and lemma 2.3.16,

$$\begin{gathered} \prod_{n=1}^{\infty} \{0,1\} = Y \cup Z \\ \Downarrow \\ \left| \prod_{n=1}^{\infty} \{0,1\} \right| = \aleph_0 \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$\begin{gathered} |[0,1)| > \aleph_0 \\ \Downarrow \\ |\mathbb{R}| > \aleph_0 \end{gathered}$$

$\square$

2.4 Metric Spaces

A function $d : X \times X \to \mathbb{R}$ is called a **metric** on X if for all $p, q, r \in X$,

(A) $d(p, q) \geq 0$

(B) $d(p, q) = 0 \iff p = q$

(C) $d(p, q) = d(q, p)$

(D) $d(p, r) \leq d(p, q) + d(q, r)$

(D) is called the **triangle inequality**. The pair (X, d) is called a **metric space**.

Theorem 2.4.1. *Let $n \in \mathbb{N}$ be given. Define $d : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ as*

$$d(\boldsymbol{x}, \boldsymbol{y}) := \|\boldsymbol{x} - \boldsymbol{y}\| = \sqrt{\sum_{k=1}^{n} (x_k - y_k)^2}$$

*for all $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$. Then d is a metric on $\mathbb{R}^n$. In particular, d is called the **standard metric** on $\mathbb{R}^n$.*

Proof. Let $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathbb{R}^n$ be given.

- $d(\boldsymbol{x}, \boldsymbol{y}) \geq 0$ is trivial.
- By Definition 1.9.1,

$$\begin{aligned} d(\boldsymbol{x}, \boldsymbol{y}) = 0 &\iff \|\boldsymbol{x} - \boldsymbol{y}\| = 0 \\ &\iff \forall k \in \mathbb{N}^n, x_k = y_k \\ &\iff \boldsymbol{x} = \boldsymbol{y} \end{aligned}$$

- $d(\boldsymbol{x}, \boldsymbol{y}) = d(\boldsymbol{y}, \boldsymbol{x})$ is trivial.
- By Theorem 1.9.2,

$$\begin{aligned} d(\boldsymbol{x}, \boldsymbol{z}) &= \|\boldsymbol{x} - \boldsymbol{z}\| = \|(\boldsymbol{x} - \boldsymbol{y}) + (\boldsymbol{y} - \boldsymbol{z})\| \\ &\leq \|\boldsymbol{x} - \boldsymbol{y}\| + \|\boldsymbol{y} - \boldsymbol{z}\| = d(\boldsymbol{x}, \boldsymbol{y}) + d(\boldsymbol{y}, \boldsymbol{z}) \end{aligned}$$

Therefore, d is a metric on $\mathbb{R}^n$. □

Definition 2.4.2. Let $k \in \mathbb{N}$ be given. A set $E \subseteq \mathbb{R}^k$ is called a k**-cell** if

$$E = \left\{ \boldsymbol{x} \in \mathbb{R}^k \;\middle|\; \forall j \in \mathbb{N}^k, a_j \leq x_j \leq b_j \right\}$$

where

$$\forall j \in \mathbb{N}^k, -\infty < a_j \leq b_j < +\infty$$

Definition 2.4.3. Let (X, d) be a metric space. Let $p \in X$ and $r \in \mathbb{R}^+$ be given.

- The **neighborhood** of p with radius r is defined as

$$N_r(p) \coloneqq \{q \in X \mid d(p, q) < r\}$$

- The **deleted neighborhood** of p with radius r is defined as

$$\widetilde{N}_r(p) \coloneqq N_r(p) \setminus \{p\} = \{q \in X \mid 0 < d(p, q) < r\}$$

Definition 2.4.4. Let (X, d) be a metric space and $E \subseteq X$. Let $p \in X$ be given.

- p is called an **limit point** of E if

$$\forall r \in \mathbb{R}^+, E \cap \widetilde{N}_r(p) \neq \varnothing$$

  The set of all limit points of E is denoted by E'.

- p is called a **isolated point** of E if

$$\exists r \in \mathbb{R}^+, E \cap N_r(p) = \{p\}$$

- p is called an **interior point** of E if

$$\exists r \in \mathbb{R}^+, N_r(p) \subseteq E$$

Definition 2.4.5. Let (X, d) be a metric space and $E \subseteq X$.

- E is said to be **open** if every point of E is an interior point of E.
- E is said to be **closed** if every limit point of E belongs to E.

Definition 2.4.6. Let (X, d) be a metric space and $E \subseteq X$.

- E is said to be **bounded** if

$$\exists p \in X, \exists r \in \mathbb{R}^+, E \subseteq N_r(p)$$

- E is said to be **perfect** if every point of E is a limit point of E.
- E is said to be **dense** if every point of X is either a limit point of E or a point of E.

Theorem 2.4.7. *Let (X, d) be a metric space.*

$$\forall p \in X, \forall r \in \mathbb{R}^+, N_r(p) \textit{ is open}$$

Proof. Let $q \in N_r(p)$ be given. Then

$$\begin{gathered} d(p,q) < r \\ \Downarrow \\ \varepsilon := r - d(p,q) > 0 \end{gathered}$$

Let $s \in N_\varepsilon(q)$ be given. Then

$$\begin{gathered} d(q,s) < \varepsilon \\ \Downarrow \\ d(p,s) \leq d(p,q) + d(q,s) < d(p,q) + \varepsilon = r \\ \Downarrow \\ s \in N_r(p) \\ \Downarrow \\ N_\varepsilon(q) \subseteq N_r(p) \end{gathered}$$

Since q is arbitrary, $N_r(p)$ is open. $\square$

Theorem 2.4.8. *Let (X, d) be a metric space and $E \subseteq X$.*

$$\forall p \in E', \forall r \in \mathbb{R}^+, |E \cap N_r(p)| \geq \aleph_0$$

Proof. Assume that

$$\exists p \in E', \exists r \in \mathbb{R}^+, |E \cap N_r(p)| < \aleph_0$$

Define

$$S := (E \cap N_r(p)) \setminus \{p\} = E \cap \widetilde{N}_r(p) \subseteq E \cap N_r(p)$$

Then

$$\begin{gathered} p \in E' \\ \Downarrow \\ S \neq \varnothing \end{gathered}$$

By Theorem 2.2.2,

$$\begin{gathered} |E \cap N_r(p)| < \aleph_0 \\ \Downarrow \\ |S| < \aleph_0 \end{gathered}$$

Define

$$T := \{d(p,q) \mid q \in S\} \neq \varnothing$$

Then

$$\begin{gathered} p \notin S \\ \Downarrow \\ 0 \notin T \end{gathered}$$

Define $f : S \twoheadrightarrow T$ as

$$f(q) = d(p, q)$$

By Theorem 2.2.4,

$$|T| < \aleph_0$$

By Theorem 2.2.5,

$$\exists \varepsilon \in \mathbb{R}^+, \varepsilon = \min T$$

Then

$$\begin{gathered} N_\varepsilon(p) \subseteq N_r(p) \\ \Downarrow \\ \forall q \in S, d(p, q) < \varepsilon \\ \Downarrow \\ E \cap \widetilde{N}_\varepsilon(p) = \varnothing \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$|E \cap N_r(p)| \geq \aleph_0$$

$\square$

Theorem 2.4.9. *Let* (X, d) *be a metric space and* $E \subseteq X$. *Then*

$$E \textit{ is open} \iff E^c \textit{ is closed}$$

Proof. ($\implies$) Let E be open. Let $p \in (E^c)'$ be given. Then

$$\forall r \in \mathbb{R}^+, E^c \cap \widetilde{N}_r(p) \neq \varnothing$$

If $p \in E$, then

$$\begin{gathered} p \in E \\ \Downarrow \\ \exists \varepsilon \in \mathbb{R}^+, N_\varepsilon(p) \subseteq E \\ \Downarrow \\ E^c \cap \widetilde{N}_\varepsilon(p) = \varnothing \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$\begin{gathered} p \notin E \\ \Downarrow \\ p \in E^c \end{gathered}$$

Since p is arbitrary,

$$\begin{gathered}(E^c)' \subseteq E^c \\ \Downarrow \\ E^c \text{ is closed}\end{gathered}$$

($\Longleftarrow$) Let E^c be closed. Let $p \in E$ be given. If

$$\nexists r \in \mathbb{R}^+, N_r(p) \subseteq E,$$

then

$$\forall r \in \mathbb{R}^+, E^c \cap N_r(p) \neq \varnothing$$

Since $p \in E$,

$$\begin{gathered}p \notin E^c \\ \Downarrow \\ E^c \cap \widetilde{N}_r(p) \neq \varnothing \\ \Downarrow \\ p \in (E^c)'\end{gathered}$$

Since E^c is closed,

$$\begin{gathered}p \in (E^c)' \\ \Downarrow \\ p \in E^c \\ \Downarrow \\ \perp\end{gathered}$$

Therefore,

$$\begin{gathered}\exists r \in \mathbb{R}^+, N_r(p) \subseteq E \\ \Downarrow \\ p \text{ is an interior point of } E\end{gathered}$$

Since p is arbitrary, E is open. $\square$

Corollary 2.4.10. *Let (X, d) be a metric space and $E \subseteq X$. Then*

$$E \textit{ is closed} \iff E^c \textit{ is open}$$

Theorem 2.4.11. *Let (X, d) be a metric space.*

$$\forall Y, (\forall y \in Y, E_y \subseteq X \textit{ is open}) \implies \bigcup_{y \in Y} E_y \textit{ is open}$$

Proof. Let $p \in \bigcup_{y \in Y} E_y$ be given. Then

$$\exists y_0 \in Y, p \in E_{y_0}$$

Since E_{y0} is open,

$$\begin{gathered}\exists r \in \mathbb{R}^+, N_r(p) \subseteq E_{y0} \\ \Downarrow \\ N_r(p) \subseteq E_{y0} \subseteq \bigcup_{y \in Y} E_y \\ \Downarrow \\ p \text{ is an interior point of } \bigcup_{y \in Y} E_y\end{gathered}$$

Since p is arbitrary, $\bigcup_{y \in Y} E_y$ is open. □

Corollary 2.4.12. *Let (X, d) be a metric space.*

$$\forall Y, (\forall y \in Y, E_y \subseteq X \textit{ is closed}) \implies \bigcap_{y \in Y} E_y \textit{ is closed}$$

Proof. By Corollary 2.4.10,

$$\forall y \in Y, (E_y)^c \text{ is open}$$

By Theorem 2.4.11,

$$\bigcup_{y \in Y} (E_y)^c \text{ is open}$$

By Theorem 2.4.9,

$$\bigcap_{y \in Y} E_y = \left(\bigcup_{y \in Y} (E_y)^c \right)^c \text{ is closed}$$

□

Theorem 2.4.13. *Let (X, d) be a metric space.*

$$(\forall k \in \mathbb{N}^n, E_k \subseteq X \textit{ is open}) \implies \bigcap_{k=1}^{n} E_k \textit{ is open}$$

Proof. Let $p \in \bigcap_{k=1}^{n} E_k$ be given. Then

$$\forall j \in \mathbb{N}^n, p \in E_j$$

Since each E_j is open,

$$\exists r_j \in \mathbb{R}^+, N_{r_j}(p) \subseteq E_j$$

for each $j \in \mathbb{N}^n$. Define

$$R := \{r_j \mid j \in \mathbb{N}^n\} \subseteq \mathbb{R}^+$$

Define $f : \mathbb{N}^n \twoheadrightarrow R$ as

$$f(j) = r_j$$

By Lemma 2.2.1 and theorem 2.2.4,

$$|R| < \aleph_0$$

By Theorem 2.2.5,

$$\exists r \in \mathbb{R}^+, r = \min R$$

Then

$$\begin{gathered} \forall j \in \mathbb{N}^n, N_r(p) \subseteq N_{r_j}(p) \subseteq E_j \\ \Downarrow \\ N_r(p) \subseteq \bigcap_{k=1}^{n} E_k \\ \Downarrow \\ p \text{ is an interior point of } \bigcap_{k=1}^{n} E_k \end{gathered}$$

Since p is arbitrary, $\bigcap_{k=1}^{n} E_k$ is open. □

Corollary 2.4.14. *Let (X, d) be a metric space.*

$$(\forall k \in \mathbb{N}^n, E_k \subseteq X \textit{ is closed}) \implies \bigcup_{k=1}^{n} E_k \textit{ is closed}$$

Proof. By Corollary 2.4.10,

$$\forall k \in \mathbb{N}^n, (E_k)^c \text{ is open}$$

By Theorem 2.4.13,

$$\bigcap_{k=1}^{n} (E_k)^c \text{ is open}$$

By Theorem 2.4.9,

$$\bigcup_{k=1}^{n} E_k = \left(\bigcap_{k=1}^{n} (E_k)^c\right)^c \text{ is closed}$$

□

Definition 2.4.15. Let (X, d) be a metric space and $E \subseteq X$. The **closure** $\overline{E}$ of E is defined as

$$\overline{E} := E \cup E'$$

Theorem 2.4.16. *Let (X, d) be a metric space and $E \subseteq X$. Then*

(A) *$\overline{E}$ is closed*

(B) *$E = \overline{E} \iff E$ is closed*

(C) *If $F \subseteq X$ satisfies*

- *$E \subseteq F$*
- *F is closed*

then $\overline{E} \subseteq F$

Proof. **(A)** Let $p \in \left(\overline{E}\right)^c$ be given. Then

$$\begin{gathered} p \notin E \wedge p \notin E' \\ \Downarrow \\ \exists r \in \mathbb{R}^+, E \cap \widetilde{N}_r(p) = \varnothing \\ \Downarrow \\ E \cap N_r(p) = \varnothing \\ \Downarrow \\ N_r(p) \subseteq E^c \end{gathered}$$

By Theorem 2.4.7, $N_r(p)$ is open. Then

$$\begin{gathered} \forall q \in N_r(p), \exists \varepsilon \in \mathbb{R}^+, N_\varepsilon(q) \subseteq N_r(p) \subseteq E^c \\ \Downarrow \\ E \cap N_\varepsilon(q) = \varnothing \\ \Downarrow \\ E \cap \widetilde{N}_\varepsilon(q) = \varnothing \\ \Downarrow \\ q \notin E' \end{gathered}$$

Since q is arbitrary,

$$\begin{gathered} N_r(p) \subseteq \left(E'\right)^c \\ \Downarrow \\ N_r(p) \subseteq \left(\overline{E}\right)^c = E^c \cap \left(E'\right)^c \\ \Downarrow \\ p \text{ is an interior point of } \left(\overline{E}\right)^c \end{gathered}$$

Since p is arbitrary,

$$\left(\overline{E}\right)^c \text{ is open}$$

By Corollary 2.4.10,

$$\overline{E} \text{ is closed}$$

(B) E is closed $\iff E' \subseteq E \iff E \cup E' = E \iff \overline{E} = E$

(C) Let $F \subseteq X$ satisfy

- $E \subseteq F$
- F is closed

Let $p \in E'$ be given. Then,

$$\begin{gathered} \forall r \in \mathbb{R}^+, E \cap \widetilde{N}_r(p) \neq \varnothing \\ \Downarrow \\ F \cap \widetilde{N}_r(p) \neq \varnothing \\ \Downarrow \\ p \in F' \subseteq F \end{gathered}$$

Since p is arbitrary,

$$\begin{gathered} E' \subseteq F' \subseteq F \\ \Downarrow \\ \overline{E} = E \cup E' \subseteq F \end{gathered}$$

$\square$

Theorem 2.4.17. *Let $(\mathbb{R}, \|\cdot\|)$ be given. If $E \subseteq \mathbb{R}$ satisfies*

- $E \neq \varnothing$
- *E is bounded above*

then

$$\exists \sup E \in \overline{E}$$

Also,

$$E \textit{ is closed} \implies \exists \sup E \in E$$

Proof. By the *least upper bound property*,

$$\exists \alpha \in \mathbb{R}, \alpha = \sup E$$

If $\alpha \in E$, then

$$\begin{gathered} E \subseteq \overline{E} \\ \Downarrow \\ \alpha \in \overline{E} \end{gathered}$$

Otherwise,

$$\begin{gathered} \forall x \in \mathbb{R}^+, \alpha - x \text{ is not an upper bound of } E \\ \Downarrow \\ \exists p \in E, \alpha - x < p < \alpha \\ \Downarrow \\ p \in \widetilde{N}_x(\alpha) \cap E \\ \Downarrow \\ \alpha \in E' \end{gathered}$$

Therefore,

$$
\begin{gathered}
E' \subseteq \overline{E} \\
\Downarrow \\
\alpha \in \overline{E}
\end{gathered}
$$

□

Theorem 2.4.18. *Let (X, d) be a metric space and $E \subseteq X$. Then*

$$E \setminus E' = \{p \in X \mid p \text{ is an isolated point of } E\}$$

Proof. **($\subseteq$)** Let $p \in E \setminus E'$ be given. Then

$$
\begin{gathered}
p \notin E' \\
\Downarrow \\
\exists r \in \mathbb{R}^{+}, E \cap \widetilde{N}_r(p) = \varnothing \\
\Downarrow \\
(E \cap N_r(p)) \setminus \{p\} = \varnothing
\end{gathered}
$$

Therefore,

$$
\begin{gathered}
p \in E \wedge p \in N_r(p) \\
\Downarrow \\
E \cap N_r(p) = \{p\} \\
\Downarrow \\
p \text{ is an isolated point of } E
\end{gathered}
$$

($\supseteq$) Let p be an isolated point of E. Then

$$\exists r \in \mathbb{R}^{+}, E \cap N_r(p) = \{p\}$$

Therefore,

$$
\begin{gathered}
p \in E \\
\Downarrow \\
E \cap \widetilde{N}_r(p) = \varnothing \\
\Downarrow \\
p \notin E' \\
\Downarrow \\
p \in E \setminus E'
\end{gathered}
$$

□

Definition 2.4.19. Let (X, d) be a metric space and $E \subseteq X$. The **boundary** ∂E of E is defined as

$$\partial E := \overline{E} \cap \overline{E^c}$$

Lemma 2.4.20. *Let* $(\mathbb{R}^n, \|\cdot\|)$ *be given for* $n \in \mathbb{N}$. *Then*

$$\forall \boldsymbol{p} \in \mathbb{R}^n, \forall r \in \mathbb{R}^+, \widetilde{N}_r(\boldsymbol{p}) \neq \varnothing$$

Proof. Let $\boldsymbol{p} \in \mathbb{R}^n$ and $r \in \mathbb{R}^+$ be given. Then

$$d\Big(\boldsymbol{p} + \frac{r}{2}\boldsymbol{e_1}, \boldsymbol{p}\Big) = \Big\|\Big(\boldsymbol{p} + \frac{r}{2}\boldsymbol{e_1}\Big) - \boldsymbol{p}\Big\| = \Big\|\frac{r}{2}\boldsymbol{e_1}\Big\| = \frac{r}{2}\|\boldsymbol{e_1}\| = \frac{r}{2} < r$$

where $\boldsymbol{e_1} = (1, 0, 0, \ldots, 0) \in \mathbb{R}^n$. Therefore,

$$\begin{gathered} \boldsymbol{p} \neq \Big(\boldsymbol{p} + \frac{r}{2}\boldsymbol{e_1}\Big) \in \widetilde{N}_r(\boldsymbol{p}) \\ \Downarrow \\ \widetilde{N}_r(\boldsymbol{p}) \neq \varnothing \end{gathered}$$

$\square$

Theorem 2.4.21. *Let* $(\mathbb{R}^n, \|\cdot\|)$ *be given for* $n \in \mathbb{N}$ *and* $E \subseteq \mathbb{R}^n$. *Then*

$$\partial E = \big(E \setminus E'\big) \cup \big(E^c \setminus (E^c)'\big) \cup \big(E' \cap (E^c)'\big)$$

This states that ∂E *consists of*

- *The set of all isolated points of* E
- *The set of all isolated points of* E^c
- *The set of all limit points of both* E *and* E^c

where the three sets are disjoint.

Proof. By definition,

$$\begin{aligned} \partial E &= \overline{E} \cap \overline{E^c} \\ &= \big(E \cup E'\big) \cap \big(E^c \cup (E^c)'\big) \\ &= (E \cap E^c) \cup \big(E \cap (E^c)'\big) \cup \big(E' \cap E^c\big) \cup \big(E' \cap (E^c)'\big) \end{aligned}$$

Consider the following equivalences:

- $E = \big(E \setminus E'\big) \cup \big(E \cap E'\big)$
- $E^c = \big(E^c \setminus (E^c)'\big) \cup \big(E^c \cap (E^c)'\big)$

Then

- $E \cap (E^c)' = \big(\big(E \setminus E'\big) \cap (E^c)'\big) \cup \big(\big(E \cap E'\big) \cap (E^c)'\big)$
- $E' \cap E^c = \big(E' \cap \big(E^c \setminus (E^c)'\big)\big) \cup \big(E' \cap \big(E^c \cap (E^c)'\big)\big)$

Consider the last two terms of the above equalities:

- $\big((E \cap E') \cap (E^c)'\big) \subseteq E' \cap (E^c)'$
- $\big(E' \cap \big(E^c \cap (E^c)'\big)\big) \subseteq E' \cap (E^c)'$

Then

$$\begin{aligned}\partial E &= \varnothing \cup \big((E \setminus E') \cap (E^c)'\big) \cup \big(E' \cap \big(E^c \setminus (E^c)'\big)\big) \cup \big(E' \cap (E^c)'\big) \\ &= \big((E \setminus E') \cap (E^c)'\big) \cup \big(E' \cap \big(E^c \setminus (E^c)'\big)\big) \cup \big(E' \cap (E^c)'\big)\end{aligned}$$

Also,

$$\begin{gathered}\forall \boldsymbol{p} \in E \setminus E', \exists r \in \mathbb{R}^+, E \cap \widetilde{N}_r(\boldsymbol{p}) = \varnothing \\ \Downarrow \\ \widetilde{N}_r(\boldsymbol{p}) \subseteq E^c\end{gathered}$$

By Lemma 2.4.20,

$$\forall s \in \mathbb{R}^+, \widetilde{N}_s(\boldsymbol{p}) \cap \widetilde{N}_r(\boldsymbol{p}) = \widetilde{N}_{\min(s,r)}(\boldsymbol{p}) \neq \varnothing$$

Then

$$\begin{gathered}\forall s \in \mathbb{R}^+, \widetilde{N}_s(\boldsymbol{p}) \cap \widetilde{N}_r(\boldsymbol{p}) \neq \varnothing \\ \Downarrow \\ \widetilde{N}_s(\boldsymbol{p}) \cap \left(\widetilde{N}_r(\boldsymbol{p}) \cap E^c\right) \neq \varnothing \\ \Downarrow \\ \varnothing \neq \left(\widetilde{N}_s(\boldsymbol{p}) \cap E^c\right) \cap \widetilde{N}_r(\boldsymbol{p}) \subseteq \widetilde{N}_s(\boldsymbol{p}) \cap E^c \\ \Downarrow \\ E^c \cap \widetilde{N}_s(\boldsymbol{p}) \neq \varnothing \\ \Downarrow \\ \boldsymbol{p} \in (E^c)'\end{gathered}$$

Then

$$\begin{gathered}E \setminus E' \subseteq (E^c)' \\ \Downarrow \\ (E \setminus E') \cap (E^c)' = E \setminus E'\end{gathered}$$

Similarly,

$$\begin{gathered}E^c \setminus (E^c)' \subseteq E' \\ \Downarrow \\ E' \cap \big(E^c \setminus (E^c)'\big) = E^c \setminus (E^c)'\end{gathered}$$

Therefore,

$$\partial E = (E \setminus E') \cup \big(E^c \setminus (E^c)'\big) \cup \big(E' \cap (E^c)'\big)$$

$\square$

Remark. Let (X, d) be a metric space and $Y \subseteq X$. Then

$$d|_{Y\times Y} : Y \times Y \to \mathbb{R}$$

satisfies the properties of a metric. Therefore, the pair

$$(Y, d|_{Y\times Y})$$

is also a metric space, which is called the **subspace** of (X, d). Since

- $d|_{Y\times Y}(p, q)$
- $(Y, d|_{Y\times Y})$

are cumbersome, we will write

- $d(p, q)$ instead of $d|_{Y\times Y}(p, q)$
- (Y, d) instead of $(Y, d|_{Y\times Y})$

for the sake of simplicity.

Definition 2.4.22. Let (X, d) be a metric space and $E \subseteq Y \subseteq X$. E is called **open relative** to Y if E is open in (Y, d). That is,

$$\forall p \in E, \exists r \in \mathbb{R}^+, Y \cap N_r(p) \subseteq E$$

Theorem 2.4.23. *Let (X, d) be a metric space and $E \subseteq Y \subseteq X$. If $V \subseteq X$ satisfies*

- *V is open in (X, d)*
- *$E = V \cap Y$*

then E is open relative to Y. The converse is also true.

Proof. **($\Longrightarrow$)** Let $V \subseteq X$ satisfies

- V is open in (X, d)
- $E = V \cap Y$

Then

$$\begin{gathered}\forall p \in E, \exists r_p \in \mathbb{R}^+, N_{r_p}(p) \subseteq V \\ \Downarrow \\ Y \cap N_{r_p}(p) \subseteq V \cap Y = E\end{gathered}$$

($\Longleftarrow$) Let E be open relative to Y. Then

$$\forall p \in E, \exists r_p \in \mathbb{R}^+, Y \cap N_{r_p}(p) \subseteq E$$

Define
$$V := \bigcup_{p \in E} N_{r_p}(p)$$
By Theorems 2.4.7 and 2.4.11, V is open in (X, d).
$$\begin{gathered} \forall p \in E, p \in N_{r_p}(p) \subseteq V \\ \Downarrow \\ E \subseteq Y \wedge E \subseteq V \\ \Downarrow \\ E \subseteq V \cap Y \end{gathered}$$
Also,
$$\begin{gathered} \forall p \in E, Y \cap N_{r_p}(p) \subseteq E \\ \Downarrow \\ \bigcup_{p \in E} \big(Y \cap N_{r_p}(p)\big) \subseteq E \\ \Downarrow \\ Y \cap \bigcup_{p \in E} N_{r_p}(p) \subseteq E \\ \Downarrow \\ Y \cap V \subseteq E \end{gathered}$$
Therefore,
$$E = V \cap Y$$
□

2.5 Compact Sets

Let (X, d) be a metric space and $E \subseteq X$. Let A be a set and let G_α be an open set in X for each $\alpha \in A$. Then the collection $\{G_\alpha\}_{\alpha \in A}$ is called an **open cover** of E if
$$E \subseteq \bigcup_{\alpha \in A} G_\alpha.$$
A collection $\{G_\alpha\}_{\alpha \in S}$ is called a **subcover** of E if

- $S \subseteq A$.
- $E \subseteq \bigcup_{\alpha \in S} G_\alpha$.

E is called **compact** if every open cover of E has a finite subcover. That is, for each open cover $\{G_\alpha\}_{\alpha \in A}$ of E, there exists a subset $S_A \subseteq A$ such that

- S_A is finite.

- $E \subseteq \bigcup_{\alpha \in S_A} G_\alpha$.

Theorem 2.5.1. *Let (X, d) be a metric space and $E \subseteq Y \subseteq X$. Then E is compact relative to Y if and only if E is compact relative to X.*

Proof. ($\implies$) Let E be a compact set relative to Y. Let $\{G_\alpha\}_{\alpha \in A}$ be an open cover of E relative to X. Then we have

$$E \subseteq \bigcup_{\alpha \in A} G_\alpha.$$

Let $H_\alpha = G_\alpha \cap Y$ for each $\alpha \in A$. By Theorem 2.4.23, H_α is open relative to Y. Since $E \subseteq Y$, we have

$$E \subseteq \left(\bigcup_{\alpha \in A} G_\alpha \right) \cap Y = \bigcup_{\alpha \in A} (G_\alpha \cap Y) = \bigcup_{\alpha \in A} H_\alpha.$$

Thus $\{H_\alpha\}_{\alpha \in A}$ is an open cover of E relative to Y. Since E is compact relative to Y, there exists a finite subcover $\{H_\alpha\}_{\alpha \in S_A}$ of E relative to Y. Then we have

$$E \subseteq \bigcup_{\alpha \in S_A} H_\alpha = \bigcup_{\alpha \in S_A} (G_\alpha \cap Y) \subseteq \bigcup_{\alpha \in S_A} G_\alpha.$$

Thus $\{G_\alpha\}_{\alpha \in S_A}$ is a finite subcover of E relative to X. Therefore, E is compact relative to X.

($\impliedby$) Let E be a compact set relative to X. Let $\{H_\alpha\}_{\alpha \in A}$ be an open cover of E relative to Y. Then we have

$$E \subseteq \bigcup_{\alpha \in A} H_\alpha.$$

By Theorem 2.4.23, there exists an open set G_α in X such that $H_\alpha = G_\alpha \cap Y$ for each $\alpha \in A$. Then we have

$$E \subseteq \bigcup_{\alpha \in A} H_\alpha = \bigcup_{\alpha \in A} (G_\alpha \cap Y) \subseteq \bigcup_{\alpha \in A} G_\alpha.$$

Thus $\{G_\alpha\}_{\alpha \in A}$ is an open cover of E relative to X. Since E is compact relative to X, there exists a finite subcover $\{G_\alpha\}_{\alpha \in S_A}$ of E relative to X. Then we have

$$E \subseteq \left(\bigcup_{\alpha \in S_A} G_\alpha \right) \cap Y = \bigcup_{\alpha \in S_A} (G_\alpha \cap Y) = \bigcup_{\alpha \in S_A} H_\alpha.$$

Thus $\{H_\alpha\}_{\alpha \in S_A}$ is a finite subcover of E relative to Y. Therefore, E is compact relative to Y. $\square$

Remark. The concept of **compactness** is preserved under the subspace topology, while the concepts of **openness** and **closedness** are not, which is the reason why

"Let (X, d) be a metric space and $E \subseteq X$."

this kind of statement is always included in theorems and definitions about metric spaces. In this textbook, you can assume that

- E is open. $\implies$ E is open relative to X.
- E is closed. $\implies$ E is closed relative to X.
- E is compact. $\implies$ E is compact relative to X.

if there is no specific mention of the subspace topology.

Theorem 2.5.2. *Let (X, d) be a metric space and $E \subseteq X$. If E is compact, then E is closed.*

Proof. Let E be compact. We will prove that E^c is open, which implies that E is closed by Corollary 2.4.10. Let $p \in E^c$ be given. Since $p \notin E$, we have

$$r_q := \frac{d(p,q)}{2} > 0 \quad \text{for each } q \in E.$$

Since $q \in N_{r_q}(q)$ for each $q \in E$, we have

$$E \subseteq \bigcup_{q \in E} N_{r_q}(q) \implies \{N_{r_q}(q)\}_{q \in E} \text{ is an open cover of } E.$$

Since E is compact, there exists a finite subcover $\{N_{r_q}(q)\}_{q \in S}$ of E. Let T be a set defined as

$$T := \{r_q \mid q \in S\}.$$

Let $f : S \to T$ be a function defined as

$$f(q) = r_q.$$

Then f is surjective. By Theorem 2.2.4, T is finite. Then, by Theorem 2.2.5, T has a minimum element, say $\varepsilon = \min T > 0$. For every $x \in N_\varepsilon(p)$, we have

$$d(x,p) < \varepsilon \leq r_q \quad \text{for each } q \in S.$$

Assume that there exists some $q \in S$ such that $x \in N_{r_q}(q)$. Then we have

$$d(p,q) \leq d(p,x) + d(x,q) < r_q + r_q = 2r_q = d(p,q),$$

which is a contradiction. Then the following statements hold:

- $N_\varepsilon(p) \cap N_{r_q}(q) = \varnothing$ for each $q \in S$.
- $E \subseteq \bigcup_{q \in S} N_{r_q}(q)$.

Therefore, we have $N_\varepsilon(p) \subseteq E^c$. Since p is arbitrary, E^c is open. □

Theorem 2.5.3. *Let (X, d) be a metric space and $E \subseteq X$. If E is compact, then E is bounded.*

Proof. Let E be compact. Let $p \in E$ be given. Let $\{G_n\}_{n \in \mathbb{N}}$ be a collection of open sets defined as

$$G_n = N_n(p) \quad \text{for each } n \in \mathbb{N}.$$

Then we have

$$n \le m \implies G_n \subseteq G_m \quad \text{for each } n, m \in \mathbb{N}.$$

For every $q \in X$, by Theorem 1.6.1, there exists $n \in \mathbb{N}$ such that $n > d(p, q)$. Thus we have

$$E \subseteq X \subseteq \bigcup_{n=1}^{\infty} G_n \implies \{G_n\}_{n \in \mathbb{N}} \text{ is an open cover of } E.$$

Since E is compact, there exists a finite subcover $\{G_n\}_{n \in S}$ of E. Since $S \subseteq \mathbb{N}$ is finite, by Theorem 2.2.5, S has a greatest element, say $m = \max S$. Then we have

$$E \subseteq \bigcup_{n \in S} G_n \subseteq G_m = N_m(p),$$

which implies that E is bounded. □

Theorem 2.5.4. *Let (X, d) be a metric space and $E \subseteq X$. If E is compact, then every closed subset $F \subseteq E$ is compact.*

Proof. Let $\{H_\alpha\}_{\alpha \in A}$ be an open cover of F. We have the following statements:

- F is closed. $\implies$ F^c is open.
- $F \subseteq \bigcup_{\alpha \in A} H_\alpha$.

Then we have

- $X = F \cup F^c \subseteq \left(\bigcup_{\alpha \in A} H_\alpha \right) \cup F^c = \bigcup_{\alpha \in A} (H_\alpha \cup F^c)$.

- $E \subseteq X \implies E \subseteq \bigcup_{\alpha \in A} (H_\alpha \cup F^c)$.

By Theorem 2.4.11, $H_\alpha \cup F^c$ is open for each $\alpha \in A$. Thus

$$\{H_\alpha \cup F^c\}_{\alpha \in A} \text{ is an open cover of } E.$$

Since E is compact, there exists a finite subcover $\{H_\alpha \cup F^c\}_{\alpha \in B}$ of E. Then we have

- $E \subseteq \bigcup_{\alpha \in B} (H_\alpha \cup F^c)$.
- $F \subseteq E \implies F \subseteq \bigcup_{\alpha \in B} (H_\alpha \cup F^c) = \left(\bigcup_{\alpha \in B} H_\alpha \right) \cup F^c$.

Since $F \cap F^c = \varnothing$, we have

$$F \subseteq \bigcup_{\alpha \in B} H_\alpha.$$

Thus $\{H_\alpha\}_{\alpha \in B}$ is a finite subcover of F. Therefore, F is compact. □

Theorem 2.5.5. *Let (X, d) be a metric space and $E \subseteq X$. If E is compact, then every infinite subset of E has a limit point in E.*

Proof. Let E be compact. Let $F \subseteq E$ be an infinite subset of E. Assume that F has no limit point in E. Then, for each $p \in E$, there exists $r_p \in \mathbb{R}^+$ such that

$$\widetilde{N}_{r_p}(p) \cap F = \varnothing \implies N_{r_p}(p) \cap F \subseteq \{p\}.$$

Let $\{G_\alpha\}_{\alpha \in E}$ be a collection of open sets defined as

$$G_\alpha = N_{r_\alpha}(\alpha) \quad \text{for each } \alpha \in E.$$

Since $\alpha \in G_\alpha$ for each $\alpha \in E$, we have

$$E \subseteq \bigcup_{\alpha \in E} G_\alpha \implies \{G_\alpha\}_{\alpha \in E} \text{ is an open cover of } E.$$

Since E is compact, there exists a finite subcover $\{G_\alpha\}_{\alpha \in S}$ of E. Then we have

$$E \subseteq \bigcup_{\alpha \in S} G_\alpha \implies F \subseteq E \subseteq \bigcup_{\alpha \in S} G_\alpha.$$

Since $F \subseteq \bigcup_{\alpha \in S} G_\alpha$, we have

$$F = F \cap \bigcup_{\alpha \in S} G_\alpha = \bigcup_{\alpha \in S} (F \cap G_\alpha) \subseteq \bigcup_{\alpha \in S} \{\alpha\} = S,$$

which is a contradiction by Theorem 2.2.2. Therefore, every infinite subset of E has a limit point in E. □

Theorem 2.5.6. *Let (X, d) be a metric space and let $\{K_\alpha\}_{\alpha \in A}$ be a collection of compact sets. Then we have*

$$\bigcap_{\alpha \in B} K_\alpha \neq \varnothing \quad \text{for all finite subset } B \subseteq A \implies \bigcap_{\alpha \in A} K_\alpha \neq \varnothing.$$

Proof. Assume that $\bigcap_{\alpha \in A} K_\alpha = \varnothing$. Then we have

$$X = \left(\bigcap_{\alpha \in A} K_\alpha \right)^c = \bigcup_{\alpha \in A} (K_\alpha)^c.$$

Since each K_α is compact, K_α is closed by Theorem 2.5.2. Then, by Corollary 2.4.10, we have

$$\text{Each } (K_\alpha)^c \text{ is open} \implies \{(K_\alpha)^c\}_{\alpha \in A} \text{ is an open cover of } X.$$

Since $K_\alpha \subseteq X \subseteq \bigcup_{\beta \in A} (K_\beta)^c$, we have

$$\{(K_\beta)^c\}_{\beta \in A} \text{ is an open cover of } K_\alpha.$$

Since K_α is compact, there exists a finite subcover $\{(K_\beta)^c\}_{\beta \in B_\alpha}$ of K_α. Then we have

$$\begin{aligned} K_\alpha \subseteq \bigcup_{\beta \in B_\alpha} (K_\beta)^c = \left(\bigcap_{\beta \in B_\alpha} K_\beta \right)^c & \\ \implies K_\alpha \cap \left(\bigcap_{\beta \in B_\alpha} K_\beta \right) &= \varnothing \\ \implies \bigcap_{\beta \in B_\alpha \cup \{\alpha\}} K_\beta &= \varnothing. \end{aligned}$$

Since $\{\alpha\}$ and B_α are finite, $B_\alpha \cup \{\alpha\}$ is also finite by Theorem 2.2.11, which contradicts the assumption. Therefore, we have

$$\bigcap_{\alpha \in A} K_\alpha \neq \varnothing.$$

□

Lemma 2.5.7. *Let $(a_n)_{n=1}^\infty$ and $(b_n)_{n=1}^\infty$ be sequences in $\mathbb{R}$, and let $\{I_n\}_{n=1}^\infty$ be a sequence of intervals in $\mathbb{R}$ such that*

- $I_n = [a_n, b_n]$ *for all* $n \in \mathbb{N}$
- $I_{n+1} \subseteq I_n \iff a_n \le a_{n+1} \le b_{n+1} \le b_n$ *for all* $n \in \mathbb{N}$

where $a_n \le b_n$ *for all* $n \in \mathbb{N}$*. Then we have*

$$\bigcap_{n=1}^{\infty} I_n \neq \varnothing.$$

Proof. Let E be a set defined as

$$E \coloneqq \{a_n \mid n \in \mathbb{N}\}.$$

Since E is bounded above by b_1, there exists $\alpha = \sup E$. Then we have

$$a_m \le \alpha \quad \text{for all } m \in \mathbb{N}.$$

For every $m \in \mathbb{N}$, we have b_m is an upper bound of E since

- $a_n \le a_m \le b_m$ for all $n \in \mathbb{N}$ such that $n \le m$.
- $a_n \le b_n \le b_m$ for all $n \in \mathbb{N}$ such that $n > m$.

Therefore, we have

$$a_m \le \alpha \le b_m \implies \alpha \in I_m \text{ for all } m \in \mathbb{N}.$$

□

Theorem 2.5.8. *Let* $k \in \mathbb{N}$ *be given. Let* $\{a_{n,j}\}_{n=1}^{\infty}$ *and* $\{b_{n,j}\}_{n=1}^{\infty}$ *be sequences in* $\mathbb{R}$ *for all* $j = 1, 2, \ldots, k$*, and let* $\{I_n\}_{n=1}^{\infty}$ *be a sequence of* k*-cells in* $\mathbb{R}^k$ *such that*

- $I_n = \left\{ \boldsymbol{x} \in \mathbb{R}^k \;\middle|\; a_{n,j} \le x_j \le b_{n,j} \text{ for all } j = 1, 2, \ldots, k \right\}$ *for all* $n \in \mathbb{N}$
- $I_{n+1} \subseteq I_n \iff a_{n,j} \le a_{n+1,j} \le b_{n+1,j} \le b_{n,j}$ *for all* $j = 1, 2, \ldots, k$ *and* $n \in \mathbb{N}$

where $a_{n,j} \le b_{n,j}$ *for all* $j = 1, 2, \ldots, k$ *and* $n \in \mathbb{N}$*. Then we have*

$$\bigcap_{n=1}^{\infty} I_n \neq \varnothing.$$

Proof. Let $\{I_{n,j}\}_{n=1}^{\infty}$ be a sequence of intervals in $\mathbb{R}$ defined as

$$I_{n,j} = [a_{n,j}, b_{n,j}] \quad \text{for all } j = 1, 2, \ldots, k \text{ and } n \in \mathbb{N}.$$

Then we have $I_{n+1,j} \subseteq I_{n,j}$ for all $j = 1, 2, \ldots, k$ and $n \in \mathbb{N}$. By Lemma 2.5.7, we have

$$\bigcap_{n=1}^{\infty} I_{n,j} \neq \varnothing \quad \text{for all } j = 1, 2, \ldots, k.$$

Let $\alpha_j \in \bigcap_{n=1}^{\infty} I_{n,j}$ be given for all $j = 1, 2, \ldots, k$. Then we have

$$\alpha_j \in I_{n,j} \implies a_{n,j} \leq \alpha_j \leq b_{n,j} \quad \text{for all } j = 1, 2, \ldots, k \text{ and } n \in \mathbb{N}.$$

Let $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \ldots, \alpha_k) \in \mathbb{R}^k$. Then we have

$$\boldsymbol{\alpha} \in I_n \quad \text{for all } n \in \mathbb{N} \implies \boldsymbol{\alpha} \in \bigcap_{n=1}^{\infty} I_n.$$

□

Theorem 2.5.9. *Let $\left(\mathbb{R}^k, \|\cdot\|\right)$ be given for $k \in \mathbb{N}$. Then every k-cell is compact.*

Proof. Let I be a k-cell in $\mathbb{R}^k$, and let $I_1 := I$. Then we have

$$I_1 = \prod_{j=1}^{k} [a_{1,j}, b_{1,j}] \subseteq \mathbb{R}^k,$$

where $-\infty < a_{1,j} \leq b_{1,j} < +\infty$ for all $j = 1, 2, \ldots, k$. Let $I_{1,j,m}$ be an interval in $\mathbb{R}$ defined as

$$I_{1,j,m} := \begin{cases} \left[a_{1,j}, \dfrac{a_{1,j} + b_{1,j}}{2}\right], & \text{if } m = 1 \\ \left[\dfrac{a_{1,j} + b_{1,j}}{2}, b_{1,j}\right], & \text{if } m = 2 \end{cases} \quad \text{for all } j = 1, 2, \ldots, k.$$

Let S_k be a set defined as

$$S_k := \{1, 2\}^k = \left\{\boldsymbol{m} \in \mathbb{R}^k \;\middle|\; m_j \in \{1, 2\} \text{ for all } j = 1, 2, \ldots, k\right\}.$$

By Theorem 2.2.13, S_k is finite. We have

$$I_1 = \bigcup_{\boldsymbol{m} \in S_k} \left(\prod_{j=1}^{k} I_{1,j,m_j}\right).$$

Let $\{G_\alpha\}_{\alpha \in A}$ be an open cover of I_1. Assume that I_1 has no finite subcover. Then, for each $\boldsymbol{m} \in S_k$, we have

$$\prod_{j=1}^{k} I_{1,j,m_j} \subseteq I_1 \subseteq \bigcup_{\alpha \in A} G_\alpha,$$

which implies that $\{G_\alpha\}_{\alpha\in A}$ is an open cover of $\prod_{j=1}^{k} I_{1,j,m_j}$. Assume that for each $\boldsymbol{m} \in S_k$, we have

$$\prod_{j=1}^{k} I_{1,j,m_j} \text{ has a finite subcover } \{G_\alpha\}_{\alpha\in A_{\boldsymbol{m}}}.$$

Then we have

$$\prod_{j=1}^{k} I_{1,j,m_j} \subseteq \bigcup_{\alpha\in A_{\boldsymbol{m}}} G_\alpha \quad \text{for all } \boldsymbol{m} \in S_k,$$

where $\bigcup_{\alpha\in A_{\boldsymbol{m}}} G_\alpha$ is open by Theorem 2.4.11. Then we have

$$I_1 \subseteq \bigcup_{\boldsymbol{m}\in S_k} \left(\bigcup_{\alpha\in A_{\boldsymbol{m}}} G_\alpha \right) = \bigcup_{\alpha\in A} G_\alpha,$$

where $A := \bigcup_{\boldsymbol{m}\in S_k} A_{\boldsymbol{m}}$ is finite by Theorem 2.2.11. Thus I_1 has a finite subcover, which is a contradiction. Therefore, there exists $\boldsymbol{m_1} \in S_k$ such that

$$\prod_{j=1}^{k} I_{1,j,(m_1)_j} \text{ has no finite subcover.}$$

Let I_2 be a k-cell in $\mathbb{R}^k$ defined as

$$I_2 := \prod_{j=1}^{k} [a_{2,j}, b_{2,j}] \subseteq \mathbb{R}^k,$$

where $a_{2,j}, b_{2,j} \in \mathbb{R}$ is defined as

- $a_{2,j} := \begin{cases} a_{1,j}, & \text{if } (m_1)_j = 1 \\ \dfrac{a_{1,j}+b_{1,j}}{2}, & \text{if } (m_1)_j = 2 \end{cases}$
- $b_{2,j} := \begin{cases} \dfrac{a_{1,j}+b_{1,j}}{2}, & \text{if } (m_1)_j = 1 \\ b_{1,j}, & \text{if } (m_1)_j = 2 \end{cases}$

for all $j = 1, 2, \ldots, k$. Then, for each $j = 1, 2, \ldots, k$, we have

$$[a_{2,j}, b_{2,j}] = I_{1,j,(m_1)_j} \subseteq [a_{1,j}, b_{1,j}],$$

which implies that

$$I_2 = \prod_{j=1}^{k} I_{1,j,(m_1)_j} \subseteq I_1.$$

Let $I_{2,j,m}$ be an interval in $\mathbb{R}$ defined as

$$I_{2,j,m} := \begin{cases} \left[a_{2,j}, \dfrac{a_{2,j}+b_{2,j}}{2}\right], & \text{if } m=1 \\ \left[\dfrac{a_{2,j}+b_{2,j}}{2}, b_{2,j}\right], & \text{if } m=2 \end{cases} \quad \text{for all } j=1,2,\dots,k.$$

Then we have

$$I_2 = \bigcup_{\boldsymbol{m}\in S_k} \left(\prod_{j=1}^{k} I_{2,j,m_j} \right).$$

We have shown that $\{G_\alpha\}_{\alpha\in A}$ is an open cover of I_2, and I_2 has no finite subcover. Then, for each $\boldsymbol{m} \in S_k$, we have

$$\prod_{j=1}^{k} I_{2,j,m_j} \subseteq I_2 \subseteq \bigcup_{\alpha\in A} G_\alpha,$$

which implies that $\{G_\alpha\}_{\alpha\in A}$ is an open cover of $\displaystyle\prod_{j=1}^{k} I_{2,j,m_j}$. Therefore, there exists $\boldsymbol{m_2} \in S_k$ such that

$$\prod_{j=1}^{k} I_{2,j,(m_2)_j} \text{ has no finite subcover.}$$

By repeating this process, we can construct the following sequences:

- The sequence $(\boldsymbol{m}_n)_{n=1}^{\infty}$ in S_k.
- The sequences $(a_{n,j})_{n=1}^{\infty}$ and $(b_{n,j})_{n=1}^{\infty}$ in $\mathbb{R}$ for all $j=1,2,\dots,k$.
- The sequence $(I_{n,j,m})_{n=1}^{\infty}$ of intervals in $\mathbb{R}$ for all $j=1,2,\dots,k$ and $m=1,2$.
- The sequence $(I_n)_{n=1}^{\infty}$ of k-cells in $\mathbb{R}^k$.

We have the following properties:

- $I_{n+1} \subseteq I_n$ for all $n \in \mathbb{N}$.
- I_n has no finite subcover for all $n \in \mathbb{N}$.
- $a_{n,j} \le a_{n+1,j} \le b_{n+1,j} \le b_{n,j}$ for all $j=1,2,\dots,k$ and $n \in \mathbb{N}$.
- $b_{n+1,j} - a_{n+1,j} = \dfrac{1}{2}(b_{n,j} - a_{n,j}) \implies b_{n,j} - a_{n,j} = \dfrac{1}{2^{n-1}}(b_{1,j} - a_{1,j})$ for all $j=1,2,\dots,k$ and $n \in \mathbb{N}$.

Let $\delta := \sqrt{\sum_{j=1}^{k}(b_{1,j} - a_{1,j})^2}$. Then we have

$$\|\boldsymbol{x} - \boldsymbol{y}\| \leq \sqrt{\sum_{j=1}^{k}(b_{n,j} - a_{n,j})^2} = \frac{\delta}{2^{n-1}} \quad \text{for all } \boldsymbol{x}, \boldsymbol{y} \in I_n \text{ and } n \in \mathbb{N}.$$

By Theorem 2.5.8, we have $\bigcap_{n=1}^{\infty} I_n \neq \varnothing$. Let $\boldsymbol{\beta} \in \bigcap_{n=1}^{\infty} I_n$ be given. Then there exists $\alpha_0 \in A$ such that $\boldsymbol{\beta} \in G_{\alpha_0}$. Since G_{α_0} is open, there exists $\varepsilon \in \mathbb{R}^+$ such that

$$B_\varepsilon(\boldsymbol{\beta}) \subseteq G_{\alpha_0}.$$

By Corollary 1.6.2, there exists $N \in \mathbb{N}$ such that $\frac{1}{2^N} < \frac{\varepsilon}{2\delta}$. Since $\boldsymbol{\beta} \in I_N$, we have

$$\|\boldsymbol{\beta} - \boldsymbol{\gamma}\| \leq \frac{\delta}{2^{N-1}} < \varepsilon \quad \text{for all } \boldsymbol{\gamma} \in I_N,$$

which implies that $I_N \subseteq B_\varepsilon(\boldsymbol{\beta}) \subseteq G_{\alpha_0}$. Therefore, I_N has a finite subcover $\{G_{\alpha_0}\}$, which is a contradiction. Thus I has a finite subcover, and I is compact. □

Corollary 2.5.10. *For every $-\infty < a \leq b < +\infty$, the closed interval $[a, b]$ is compact in $(\mathbb{R}, \|\cdot\|)$.*

Theorem 2.5.11. *Let $(\mathbb{R}^n, \|\cdot\|)$ be given for $n \in \mathbb{N}$ and $E \subseteq \mathbb{R}^n$. Then we have*

$$E \text{ is compact} \iff E \text{ is closed and bounded.}$$

*This theorem is called the **Heine-Borel Theorem**.*

Proof. ($\Longrightarrow$) Let E be compact. E is closed by Theorem 2.5.2, and E is bounded by Theorem 2.5.3.

($\Longleftarrow$) Let E be closed and bounded. Then there exists some $\boldsymbol{p} \in \mathbb{R}^n$ and $r \in \mathbb{R}^+$ such that $E \subseteq B_r(\boldsymbol{p})$. Let I be a n-cell in $\mathbb{R}^n$ defined as

$$I := \prod_{j=1}^{n}[p_j - r, p_j + r] \subseteq \mathbb{R}^n.$$

Then we have $E \subseteq B_r(\boldsymbol{p}) \subseteq I$. By Theorem 2.5.9, I is compact. Since E is closed, by Theorem 2.5.4, E is compact. □

2.6 Connected Sets

Let (X, d) be a metric space and $E \subseteq X$. E is called **connected** if there exists no pair of sets $A, B \subseteq X$ such that

- $\varnothing \neq A \subseteq E$ and $\varnothing \neq B \subseteq E$.
- $\overline{A} \cap B = A \cap \overline{B} = \varnothing$.
- $E = A \cup B$.

Such pair of sets A and B is called **separated**. If E is not connected, then E is called **disconnected**.

Theorem 2.6.1. *Let $(\mathbb{R}, \|\cdot\|)$ be given. Then $E \subseteq \mathbb{R}$ is connected if and only if*

$$a < x < b \implies x \in E \quad \textit{for all } a, b \in E \textit{ and } x \in \mathbb{R}.$$

Proof. ($\implies$) Let $E \subseteq \mathbb{R}$ be connected. Assume that there exist $a, b \in E$ and $x \in \mathbb{R}$ such that

$$a < x < b \text{ and } x \notin E.$$

Let A, B be sets defined as

- $A := E \cap (-\infty, x) \subseteq E$.
- $B := E \cap (x, \infty) \subseteq E$.

Then we have

- $A \neq \varnothing$ and $B \neq \varnothing$ since $a \in A$ and $b \in B$.
- $\overline{A} \cap B = A \cap \overline{B} = \varnothing$.
- $E = A \cup B$ since $x \notin E$.

Thus A and B are separated, which is a contradiction. Therefore, we have

$$a < x < b \implies x \in E \quad \text{for all } a, b \in E \text{ and } x \in \mathbb{R}.$$

($\impliedby$) Let $E \subseteq \mathbb{R}$ be a set such that

$$a < x < b \implies x \in E \quad \text{for all } a, b \in E \text{ and } x \in \mathbb{R}.$$

Assume that E is disconnected. Then there exist separated sets $A, B \subseteq \mathbb{R}$ such that

- $\varnothing \neq A \subseteq E$ and $\varnothing \neq B \subseteq E$.

- $\overline{A} \cap B = A \cap \overline{B} = \varnothing$.
- $E = A \cup B$.

Let $a \in A$ and $b \in B$ be given. Without loss of generality, assume that $a < b$. Let

$$\alpha := \sup(A \cap [a, b]) \in \mathbb{R},$$

where we have

- $a \in A \implies A \cap [a, b] \neq \varnothing$.
- $A \cap [a, b]$ is bounded above by b.

Then the *existence* of α is guaranteed by the *least upper bound property*. By Theorem 2.4.17, we have

$$\alpha \in \overline{(A \cap [a, b])} = (A \cap [a, b]) \cup (A \cap [a, b])'.$$

By Definition 2.4.4, it is clear that

$$(A \cap [a, b])' \subseteq A'.$$

Thus we have

$$\alpha \in (A \cap [a, b]) \subseteq A \quad \text{or} \quad \alpha \in (A \cap [a, b])' \subseteq A' \implies \alpha \in \overline{A}.$$

Then the following statements hold:

- $\alpha = \sup(A \cap [a, b]) \implies a \leq \alpha \leq b$.
- $\alpha \in \overline{A} \implies \alpha \notin B \implies \alpha \neq b$.

If $\alpha \notin A$, then we have

- $\alpha \neq a \implies a < \alpha < b$
- $\alpha \notin A$ and $\alpha \notin B \implies \alpha \notin E$

which is a contradiction. Otherwise, we have

$$\alpha \in A \implies \alpha \notin \overline{B}.$$

Then there exists $r \in (0, b - \alpha)$ such that

$$\widetilde{N}_r(\alpha) \cap B = \varnothing \implies \beta := \alpha + r \notin B.$$

Then we have

- $a \leq \alpha < \beta < b$
- $\beta > \alpha \implies \beta \notin A \implies \beta \notin E$

which is a contradiction. Therefore, we conclude that E is connected. $\square$

Chapter 3

Sequences and Series

3.1 Convergent Sequences

Let (X, d) be a metric space. A **sequence** $(x_n)_{n=1}^{\infty}$ is a function $f : \mathbb{N} \to X$, where $x_n = f(n)$ for each $n \in \mathbb{N}$.

A sequence $(x_n)_{n=1}^{\infty}$ is said to be **convergent** if there exists $x \in X$ such that

$$\text{For every } \varepsilon \in \mathbb{R}^+, \text{ there exists } N \in \mathbb{N} \text{ such that } n \geq N \implies d(x_n, x) < \varepsilon.$$

In this case, we say that

- $(x_n)_{n=1}^{\infty}$ **converges** to x.
- x is the **limit** of $(x_n)_{n=1}^{\infty}$.

We denote the limit of $(x_n)_{n=1}^{\infty}$ by

- $\lim\limits_{n\to\infty} x_n = x$.
- $x_n \to x$ as $n \to \infty$.

If $(x_n)_{n=1}^{\infty}$ is not convergent, it is said to be **divergent**.

Example 3.1.1. Let $(\mathbb{R}, \|\cdot\|)$ be given. Let $(x_n)_{n=1}^{\infty}$ be a sequence defined as

$$x_n = \frac{1}{n} \quad \text{for each } n \in \mathbb{N}.$$

Then, for each $\varepsilon \in \mathbb{R}^+$, we have

$$n > \frac{1}{\varepsilon} \implies \|x_n - 0\| = \left|\frac{1}{n} - 0\right| = \frac{1}{n} < \varepsilon.$$

Therefore, $(x_n)_{n=1}^{\infty}$ converges to 0.

Let $\left(\mathbb{R}^+, \|\cdot\|\right)$ be a subspace of $(\mathbb{R}, \|\cdot\|)$. Let $(x_n)_{n=1}^\infty$ be a sequence defined as

$$x_n = \frac{1}{n} \quad \text{for each } n \in \mathbb{N}.$$

By Theorem 1.6.1, for each $y \in \mathbb{R}^+$, there exists $N_y \in \mathbb{N}$ such that $N_y > \frac{1}{y}$. Let

$$\varepsilon_y := y - \frac{1}{N_y} \in \mathbb{R}^+ \quad \text{for each } y \in \mathbb{R}^+.$$

Then we have

$$n > N_y \implies \|x_n - y\| = \left|\frac{1}{n} - y\right| > \left|\frac{1}{N_y} - y\right| = \varepsilon_y.$$

Therefore, $(x_n)_{n=1}^\infty$ diverges in $\left(\mathbb{R}^+, \|\cdot\|\right)$.

Remark. Let (X, d) be a metric space. In this textbook, $(x_n)_{n=1}^\infty$ represents a sequence in (X, d), while $\{x_n\}_{n=1}^\infty$ represents a set defined as

$$\{x_n\}_{n=1}^\infty := \{x_n \mid n \in \mathbb{N}\} \subseteq X.$$

Definition 3.1.2. Let (X, d) be a metric space. A sequence $(x_n)_{n=1}^\infty$ is said to be **Bounded** if $\{x_n\}_{n=1}^\infty$ is bounded.

Theorem 3.1.3. *Let (X, d) be a metric space. Let $(x_n)_{n=1}^\infty$ be a convergent sequence. Then we have*

- ***(A)*** *$(x_n)_{n=1}^\infty$ is bounded.*
- ***(B)*** *The limit of $(x_n)_{n=1}^\infty$ is unique.*

Proof. Let x be the limit of $(x_n)_{n=1}^\infty$.

(A) Let $\varepsilon = 1$. Then there exists $N \in \mathbb{N}$ such that

$$n \geq N \implies d(x_n, x) < 1.$$

Let S be a set defined as

$$S := \{d(x_n, x) \mid 1 \leq n \leq N\} \subseteq [0, \infty).$$

Let $f : \mathbb{N}^N \to S$ be a function defined as

$$f(k) = d(x_k, x) \quad \text{for all } k \in \mathbb{N}^N.$$

Then f is surjective. By Lemma 2.2.3, S is finite. Then, by Theorem 2.2.5, S has a greatest element, say $M = \max S$. Then we have

$$d(x_n, x) < M + 1 \quad \text{for all } n \in \mathbb{N}.$$

Therefore, $(x_n)_{n=1}^\infty$ is bounded.

(B) Let y be the limit of $(x_n)_{n=1}^{\infty}$. Assume that $x \neq y$. Then we have $r := d(x, y) > 0$. Let $\varepsilon = \dfrac{r}{2}$. Then we have

- There exists $N_x \in \mathbb{N}$ such that $n \geq N_x \implies d(x_n, x) < \dfrac{r}{2}$.
- There exists $N_y \in \mathbb{N}$ such that $n \geq N_y \implies d(x_n, y) < \dfrac{r}{2}$.

Thus we have

$$n \geq \max\{N_x, N_y\} \implies d(x, y) \leq d(x, x_n) + d(x_n, y) < r,$$

which is a contradiction. Therefore, we have $x = y$.

□

Theorem 3.1.4. *Let (X, d) be a metric space. Then the following statements hold:*

(A) *A sequence $(x_n)_{n=1}^{\infty}$ converges to $x \in X$ if and only if*

For every $\varepsilon \in \mathbb{R}^+$, $N_\varepsilon(x)$ contains all but finitely many terms of $(x_n)_{n=1}^{\infty}$.

(B) *Let $E \subseteq X$ be given. If x is a limit point of E, then there exists a sequence $(x_n)_{n=1}^{\infty}$ in (X, d) such that*

$$\lim_{n \to \infty} x_n = x.$$

Proof. **(A)** **($\implies$)** Let $(x_n)_{n=1}^{\infty}$ converge to $x \in X$. Then, for each $\varepsilon \in \mathbb{R}^+$, there exists $M_\varepsilon \in \mathbb{N}$ such that

$$n \geq M_\varepsilon \implies d(x_n, x) < \varepsilon,$$

which implies that

$$\{x_n\}_{n=M_\varepsilon}^{\infty} \subseteq N_\varepsilon(x).$$

($\impliedby$) Let $(x_n)_{n=1}^{\infty}$ be a sequence in (X, d) such that

For every $\varepsilon \in \mathbb{R}^+$, $N_\varepsilon(x)$ contains all but finitely many terms of $(x_n)_{n=1}^{\infty}$.

(B) Let $E \subseteq X$ be given. Assume that x is a limit point of E. Then for every $\varepsilon \in \mathbb{R}^+$, we have $(N_\varepsilon(x) - \{x\}) \cap E \neq \varnothing$. Thus we can choose a sequence $(y_n)_{n=1}^{\infty}$ in (X, d) such that

$$y_n \in (N_{\frac{1}{n}}(x) \setminus \{x\}) \cap E.$$

Then we have

$$d(y_n, x) < \frac{1}{n} \quad \text{for each } n \in \mathbb{N},$$

which implies that $\lim_{n \to \infty} y_n = x$.

□

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10.9 Vector Analysis

Chapter 11

The Lebesgue Theory

11.1 Set Functions

11.2 Construction of the Lebesgue Measure

11.3 Measure Spaces

11.4 Measurable Functions

11.5 Simple Functions

11.6 Integration

11.7 Comparison with the Riemann Integral

11.8 Integration of Complex Functions

11.9 Functions of Class $\mathscr{L}^2$

Appendix: Construction of $\mathbb{R}$

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